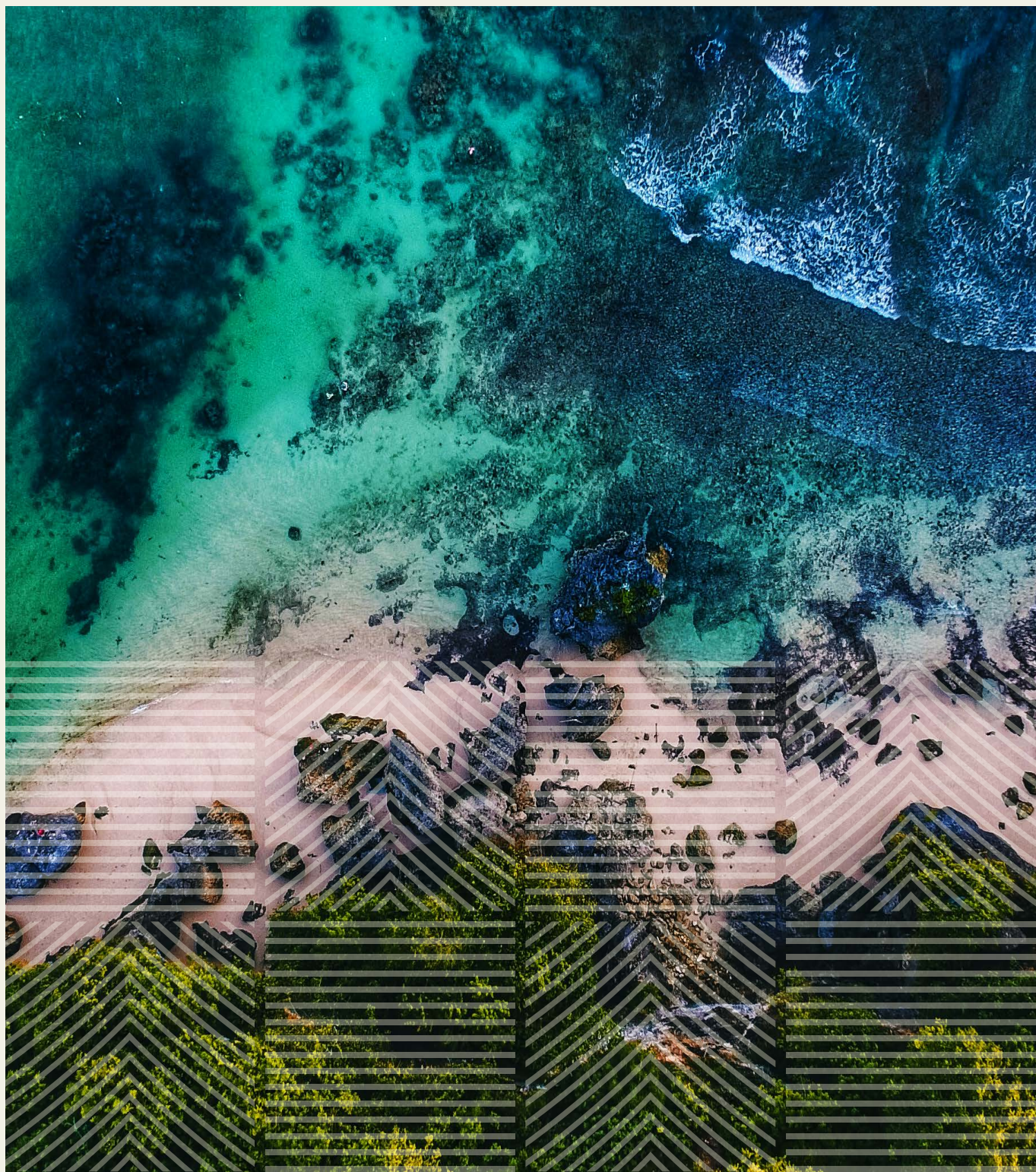


08/2021

Monetary Policy **STATEMENT**



Reserve Bank
of New Zealand
Te Pūtea Matua





STATEMENT

of the MPC's monetary policy strategy

The Monetary Policy Committee's (MPC) monetary policy strategy is its overarching plan for how it will formulate monetary policy under different circumstances to achieve its objectives.¹ It outlines a consistent approach to how the MPC intends to achieve its objectives across time, accounting for trade-offs and uncertainty. Agreeing on and publishing a strategy promotes transparency, public understanding, and accountability.

Monetary policy framework and objectives

Under the *Reserve Bank of New Zealand Act 1989* (the Act), the MPC is responsible for formulating monetary policy to maintain a stable general level of prices over the medium term and to support maximum sustainable employment.² Operational objectives for monetary policy are set out in the *Remit*. The current *Remit* sets out a flexible inflation targeting regime, under which the MPC must set policy to:

- keep future annual inflation between 1 and 3 percent over the medium term, with a focus on keeping future inflation near the 2 percent mid-point; and
- support maximum sustainable employment, considering a broad range of labour market indicators and taking into account that maximum sustainable employment is largely determined by non-monetary factors.

In pursuing these objectives, the *Remit* requires the MPC to have regard to the efficiency and soundness of the financial system, seek to avoid unnecessary instability in the economy and financial markets, and discount events that have only transitory effects on inflation. The MPC must also assess the effect of its monetary policy decisions on the Government's policy to support more sustainable house prices.

The Reserve Bank's flexible inflation targeting framework and the MPC's monetary policy strategy reflect the fact that:

- low and stable inflation is monetary policy's best long-run contribution to the well-being of New Zealanders;
- in the short to medium term, monetary policy can influence real variables such as employment, and hence policy trade-offs can arise; and

- monetary policy is more effective if the Bank's policy targets are credible, so policy should be formulated in a way that ensures credibility is maintained.

Key aspects of monetary policy strategy

The MPC practises **forecast targeting**, which means that it sets monetary policy such that it expects to achieve its inflation and employment goals in the medium term. In most instances the MPC aims to return inflation to the target mid-point within a one to three year horizon. The appropriate horizon at each policy decision will vary based on how different policy paths will contribute to maximum sustainable employment, whether price-setters' expectations are consistent with the inflation target, and other considerations such as the balance of risks to the MPC's central economic outlook.

¹ For a more in-depth discussion of monetary policy strategy in New Zealand, see J. Ratcliffe and R. Kendall (2019), '[Monetary policy strategy in New Zealand](#)', Reserve Bank of New Zealand, *Bulletin*, Vol. 82, No. 3, April.

² These economic objectives contribute to the overall purpose of the Act, which is to promote the prosperity and well-being of New Zealanders, and contribute to a sustainable and productive economy. See [monetary policy framework](#) for more information on New Zealand's monetary policy framework, including the full text of the *Remit*.

The MPC does not attempt to return inflation and employment to target immediately, because monetary policy actions take time to transmit through the economy. Attempting to return inflation to target too quickly would result in unnecessary instability in the economy and financial markets. The 1 to 3 percent target range for inflation provides the MPC with flexibility to ensure that managing inflation variability does not come at the cost of excessive variability in the real economy. For similar reasons, the MPC does not attempt to offset events that are expected to have only transitory effects on inflation.

The MPC **takes into account both its inflation and employment objectives** when setting policy. In the long run, no trade-off exists between the MPC's objectives. In the short to medium term, there may be situations where monetary policy can move one objective closer to target only at the cost of the other, resulting in a trade-off. When a trade-off does arise, the MPC will consider outcomes for both objectives in setting policy. In general, if employment is projected to be below its long-run sustainable level, the MPC would let inflation overshoot the target mid-point for a time, and vice versa (while staying within the 1–3 percent target range).

The MPC **responds to both deviations above target and deviations below target**. The MPC sets policy to stabilise employment near its maximum sustainable level, and to return inflation near to the target mid-point, regardless of whether inflation is currently below or above 2 percent. This approach helps to anchor inflation expectations at the target mid-point and promotes sustainable growth and employment by dampening fluctuations in the business cycle.

The MPC **considers the balance of risks** to its objectives that arise from uncertainty about the economic outlook and the transmission of its policy decisions. In general, the MPC will incorporate likely future developments into its central economic projections and set monetary policy in response. However, the MPC will also take into account risks to its central projections when setting policy. Under extreme uncertainty, the MPC may choose to publish scenarios instead of central projections to illustrate the range of possible situations and economic outcomes that could occur when circumstances are highly unpredictable.

The MPC **has regard to the efficiency and soundness of the financial system**, while recognising that in most instances prudential policy is better suited to leaning against risks to financial stability. The Reserve Bank takes prudential policy settings into account when setting monetary policy, and vice versa.

Implementation of strategy

The MPC applies the following process when formulating a policy decision:

1. Firstly, it assesses the outlook for the economy and the implications for its policy objectives. It then discusses risks to achieving its policy objectives.
2. Next, it considers which stance of monetary policy is most consistent with its monetary policy strategy given the current economic outlook, risks, and trade-offs.
3. Finally, the MPC decides how it will achieve the desired stance of monetary policy, including whether or not to adjust its policy settings at the current meeting and how it will communicate the policy outlook. The MPC has a **suite of monetary tools** to achieve its goals, and uses its **Principles for Monetary Tools** to make decisions on which tools to deploy.



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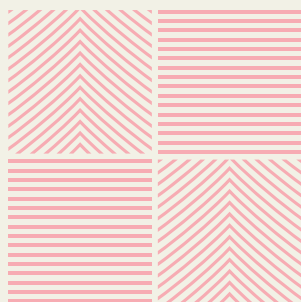
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Monetary Policy **STATEMENT**

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The projection was finalised on 18 August 2021. The OCR projection incorporates an outlook for monetary policy that is consistent with the MPC's monetary policy assessment, which was finalised on 18 August 2021. Other data and developments have been incorporated up until 11 August 2021.



Policy
assessment

CHAPTER 01

CHAPTER 1

Policy assessment

Tēnā koutou katoa, welcome all.

The Monetary Policy Committee agreed to retain the current stimulatory level of monetary settings, keeping the Official Cash Rate (OCR) at 0.25 per cent for now. Today's decision was made in the context of the Government's imposition of Level 4 COVID restrictions on activity across New Zealand.

The Committee will assess the inflation and employment outlook on an ongoing basis, with a view to continue to reduce the level of monetary stimulus over time so as to best meet their policy remit. This follows the recent halting of additional government bond purchases under the Large Scale Asset Purchase (LSAP) programme in July.

Global monetary and fiscal settings remain at accommodative levels, supporting international spending and investment. Rising vaccination rates across many countries have provided economic impetus. The rise in activity has continued to support demand and prices for New Zealand's export commodities.

However, the need to reinstate COVID-19 containment measures in some regions highlights the serious health and economic risks posed by the virus. Persistent and elevated health risks are promoting ongoing global supply chain disruptions, and are acting to constrain productive capacity and prolong inflationary pressures. Today's re-introduction of Level 4 restrictions to activity across New Zealand is a stark example of how unpredictable and disruptive the virus is proving to be.

The Committee noted that the New Zealand economy had rebounded more strongly than most countries, with less domestic disruption caused by COVID-19 to date. Employment is currently at or above its maximum sustainable level, and consumer price inflation expectations remain anchored near 2 percent, the midpoint of the target range.

Recent data for the New Zealand economy suggest demand is robust and the economic recovery has broadened, despite some weakness persisting in the sectors most exposed to international tourism. Household spending and construction activity are at high levels and continue to grow, and business investment is responding to increased demand.

Capacity pressures are now evident in the economy, particularly in the labour market where job vacancies remain high despite the recent decline in unemployment and underemployment. Wages are rising consistent with the tight labour market conditions.

Broader inflation pressures are being accentuated in the near term by one-off price rises such as higher oil prices, and temporary factors such as supply shortfalls and higher transport costs. Near-term consumer price inflation is expected to rise above the Committee's target range before returning towards the 2 percent midpoint around mid-2022.

The Committee agreed they are confident of meeting their inflation and employment remit with less need for the existing level of monetary stimulus. However, the Committee remains alert to the supply disruptions that COVID-19 can create, and the dampening effect this can have on confidence. House prices are also above their sustainable level, heightening the risk of a price correction as supply increases.

The Committee agreed that their least regrets policy stance is to further reduce the level of monetary stimulus so as to anchor inflation expectations and continue to contribute to maximum sustainable employment. They agreed, however, to keep the OCR unchanged at this meeting given the heightened uncertainty with the country in a lockdown.

Meitaki, thanks.



Adrian Orr
Governor



SUMMARY

Record of meeting

The Monetary Policy Committee discussed economic developments since the May *Statement*. The Committee noted that the global economy has continued to recover, supported by rising vaccination rates in many countries, a gradual relaxation of mobility restrictions, and continued monetary and fiscal support.

The Monetary Policy Committee discussed economic developments since the May *Statement*. The Committee noted that the global economy has continued to recover, supported by rising vaccination rates in many countries, a gradual relaxation of mobility restrictions, and continued monetary and fiscal support.

The Committee noted the considerable uncertainty that exists regarding the longer-run impacts of COVID-19, particularly with the emergence of new variants. Globally, periods of health-related mobility restriction are likely to continue for some time, creating ongoing short-term economic disruptions, supply cost pressures, and lower productive capacity.

The Committee agreed that in New Zealand the recent economic data suggest domestic demand is robust and that the economic recovery has broadened in recent months. While weakness still persists in sectors most heavily exposed to international tourism, activity in most industries now exceeds pre-COVID levels.

Domestic economic activity has been underpinned by strong household spending, high levels of construction, and strong demand for New Zealand's commodity exports. Recent

data has also shown a pick-up in business investment, which broadens the base of aggregate demand and suggests businesses are responding to emerging capacity constraints.

The Committee noted uncertainty related to the emergence of new cases of COVID-19 in the community and the move back into Alert Level 4. The reinstatement of the Government Wage Subsidy Scheme and COVID-19 Resurgence Support Payments is expected to significantly buffer the loss of income associated with the lockdown.

The Committee agreed that capacity constraints were building in the economy. Pressures are particularly acute in the labour market, where job vacancies remain high alongside declines in unemployment. Falling underemployment provides a greater level of confidence that spare capacity is being absorbed. Employment is assessed as being at or above its maximum sustainable level in the current environment.

Wage inflation has increased in line with the tightening in the labour market, but the Committee expressed uncertainty about whether higher wage growth will be sustained.

The Committee noted that capacity constraints are contributing to rising headline inflation. Mirroring global developments, inflationary pressure in New Zealand has been accentuated in the near term by one-off factors such as higher oil prices, and temporary factors such as supply shortfalls and rising transport costs. This is expected to push inflation above 4 percent in the near term, before returning towards the 2 percent midpoint of the target band from mid-2022. Medium and long-term inflation expectations remain anchored at 2 percent.

The Committee reflected that experience over the past 12 months has provided more confidence about the resilience of domestic demand in the face of health-related restrictions. The Government Wage Subsidy proved effective in supporting domestic incomes and providing job security through periods of lockdown, which has enabled a rapid recovery in consumer spending. This scheme has been rapidly reinstated in light of the current lockdown. While some households suffered income losses and accumulated debt, many households retain a larger buffer of savings, which could provide ongoing support to consumption.

The Committee acknowledged that restrictions on the movement of people across the New Zealand border will only be removed gradually, and subject to ongoing health-related uncertainty. However, they also agreed that, to date, increased domestic spending has provided a significant offset to the loss of international tourism earnings. The closure of the border has also reduced international labour mobility, creating capacity shortages in some industries that have traditionally been reliant on migrant labour.

In light of this experience, members expressed caution about the level of remaining supply capacity in the New Zealand economy. The economic disruption caused by the ongoing global health issues has increased skill mismatches, which has likely reduced maximum sustainable employment in the near term. The Committee discussed the risk that the productive capacity of the economy is lagging domestic demand, which could lead to more persistent inflation pressure.

The Committee discussed the current, and risk of future, outbreaks of COVID-19 in New Zealand, and how monetary policy should respond. The Committee agreed that fiscal policy (government spending and transfer payments) has proved to be a very effective tool to respond to any immediate reduction in demand in the event of outbreaks. A monetary policy response may be required if a health-related lockdown has a more enduring impact on inflation and employment.

As required by their Remit, members assessed the impact of monetary policy on the Government's objective to support more sustainable house prices. The Committee noted the Reserve

Bank's assessment that the level of house prices is currently unsustainable. Members noted that the Reserve Bank is currently consulting on further bank lending restrictions to help mitigate the financial stability risks associated with unsustainable house prices.

The Committee noted that a number of factors are expected to weigh on house prices over the medium term. These include strong house building, slower population growth, changes to tax settings, and the ongoing impacts of tighter bank lending rules. Rising mortgage interest rates, as monetary stimulus is reduced, would also constrain house prices to a more sustainable level. Members expressed uncertainty about how quickly momentum in the housing market will recede and noted a risk that any continued near-term price growth could lead to sharper falls in house prices in the future.

The Committee reiterated that the OCR is currently the preferred tool to adjust the level of stimulus in the economy. The principles governing the suite of monetary policy tools will continue to guide their use. In line with those principles, the Funding for Lending Programme (FLP) will remain in place under its current terms until the drawdown window expires next year. The Committee directed staff to develop an operational strategy to help inform decisions regarding the management of government and Local Government Funding Agency (LGFA) bonds purchased under the Large Scale Asset Purchase (LSAP) programme, consistent with the Committee's desired stance of policy and supporting the functioning of markets.

The Committee discussed the stance of monetary policy. Members noted that they now had more confidence that rising

capacity pressures will feed through into inflation, and that employment is at its maximum sustainable level. Members concluded that they could continue removing monetary stimulus, following their decision to halt additional purchases of government bonds under the LSAP programme at their July meeting.

The Committee discussed the merits of an increase in the OCR at this meeting and considered the implications of alternative sequencing of OCR changes over time. The Committee agreed that their least regrets policy stance is to further reduce monetary policy stimulus to reduce the risk that inflation expectations become unanchored. However, in light of the current Level 4 lockdown and health uncertainty the Committee agreed to leave the OCR unchanged at this meeting.

On Wednesday 18 August, the Committee reached a consensus to:

- Maintain the OCR at 0.25 percent;
- Direct staff to develop an operational strategy to inform decisions on the management of government and LGFA bonds purchased under the LSAP programme; and
- Maintain the existing Funding for Lending Programme conditions.

Attendees:

Reserve Bank members of MPC:

Adrian Orr, Geoff Bascand, Christian Hawkesby, Yuong Ha

External MPC members:

Bob Buckle, Peter Harris, Caroline Saunders

Treasury Observer:

Bryan Chapple

MPC Secretary:

Chris Bloor



Current economic
assessment
and risks

CHAPTER 02

CHAPTER 2

Current economic assessment and risks

The Reserve Bank's Monetary Policy Committee (MPC) is responsible for setting monetary policy to achieve and maintain medium-term price stability, and support maximum sustainable employment. These outcomes are the best long-run contribution monetary policy can make to the prosperity and well-being of New Zealanders.



Key points

- Economic growth in New Zealand has been stronger and more broad based than previously anticipated, with activity in most sectors of the economy having returned to pre-COVID-19 levels. While the recent move to Alert Level 4 will restrain activity, the impact will depend on the length of time restrictions stay in place and the responses of households and businesses.
- Much of the economic momentum in recent months has been driven by strength in spending by households and continued strong demand for our commodity exports. Businesses are also showing increased confidence to invest and hire more workers.
- The pressure on available resources in the economy is now much greater than was assumed in the May *Statement*. Pressures are particularly acute in the labour market, where job vacancies remain high despite declines in unemployment. Employment is assessed as being at or above its maximum sustainable level in the current environment.
- When combined with supply-chain disruptions and increases in global oil prices, capacity pressures in the economy have resulted in annual CPI inflation increasing to 3.3 percent. Inflation is expected to temporarily increase further in the near term, and return to the mid-point of the 1 to 3 percent target band over the next three years.
- It is unclear whether recent and expected high inflation will become embedded in price- and wage-setting behaviour, which could lead to sustained inflationary pressures.
- Other risks to the economic outlook remain. The emergence of more contagious variants of COVID-19 illustrates the vulnerability of the economic recovery to further outbreaks. The timing and economic effect of easing border restrictions are also uncertain, as are the longer-term impacts of COVID-19 on productive capacity here and abroad.
- Emergency levels of monetary stimulus are no longer warranted given the strength in employment and rising inflationary pressures. Our central projection implies that monetary stimulus should be reduced to ensure the MPC meets its objectives.

Current economic assessment

Economic growth has been stronger than expected

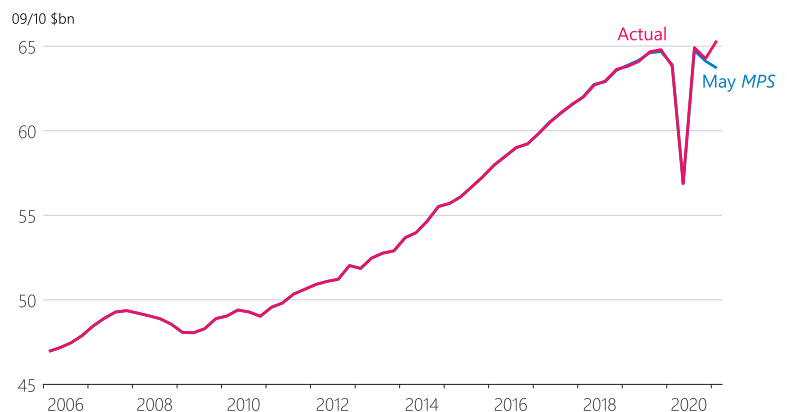
Economic growth in New Zealand has been stronger and more broad based than previously expected (figure 2.1). The economy expanded 1.6 percent in the first three months of 2021, despite a lack of international visitors during what is usually the peak tourism season (figure 2.2). While activity in the retail, healthcare and construction sectors has been particularly strong, activity in most sectors has been around pre-COVID-19 levels – with the exception of mining and those most exposed to the loss in international visitors (figure 2.3).

Household spending is underpinning much of the momentum

Much of the momentum in economic growth has been due to strength in household spending, and demand for our goods exports. Household consumption increased 5.4 percent in the March 2021 quarter, taking it above pre-COVID-19 levels despite very low population growth (figure 2.4). New Zealanders appear to have redirected much of their usual spending on overseas holidays towards travelling within New Zealand and purchases of durable goods.

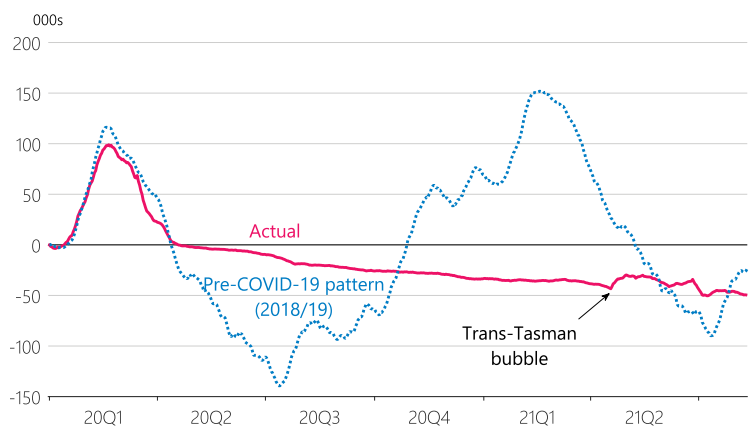
Interpretation of the most recent consumption data is complicated by the impact of border restrictions on households' spending behaviour, such as when and where they choose to go on holiday. Reflecting this, measures of consumption that attempt to account for usual seasonal variation in spending are likely to remain volatile.

Figure 2.1
Quarterly production GDP
(seasonally adjusted)



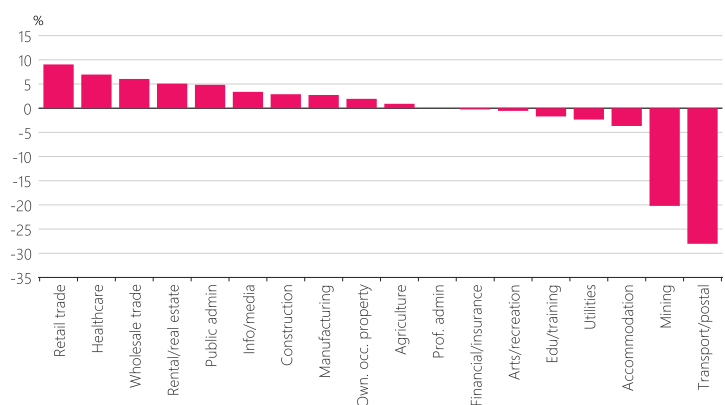
Source: Stats NZ, RBNZ estimates.

Figure 2.2
Net New Zealand border crossings
(cumulative, all countries)



Source: New Zealand Customs Service, RBNZ estimates.

Figure 2.3
Production GDP by industry
(comparison to 2019Q4, seasonally adjusted)

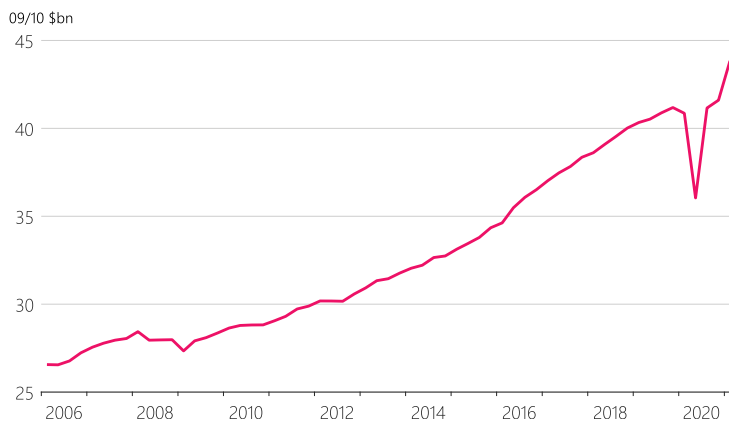


Source: Stats NZ, RBNZ estimates.

Note: Unallocated taxes are not shown.

Figure 2.4

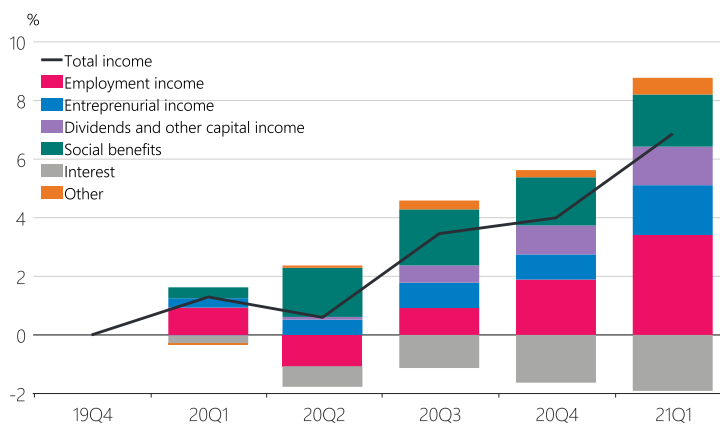
Quarterly household consumption expenditure (seasonally adjusted)



Source: Stats NZ.

Figure 2.5

Nominal household income growth (cumulative change since 2019Q4, seasonally adjusted)

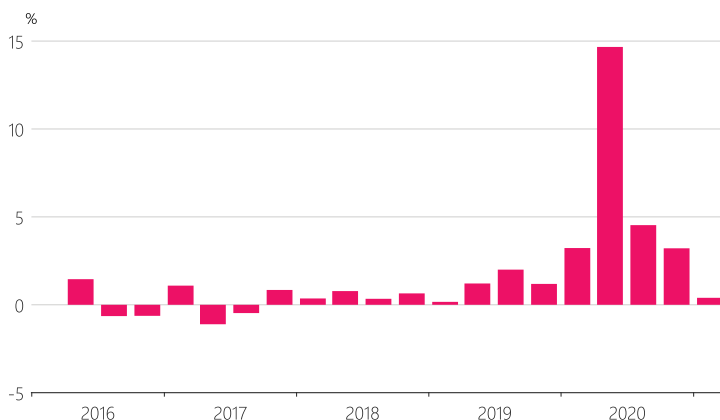


Source: Stats NZ, RBNZ estimates.

Note: Chart shows the change in quarterly nominal household income since the December 2019 quarter.

Figure 2.6

Household saving rate (seasonally adjusted)



Source: Stats NZ.

Note: The household saving rate is defined as net household saving as a share of net household disposable income.

However, we do consider there to be strong underlying momentum in household spending, consistent with higher employment, strong growth in household incomes, savings accumulated during 2020, and high house price inflation. Newly available national accounts data show that aggregate nominal household sector income was above pre-COVID-19 levels in the March 2021 quarter. Household incomes have increased significantly since the outset of the COVID-19 pandemic. This reflects recovering employment – particularly in more recent quarters – and social support payments that remain higher than previously (figure 2.5).

Savings accumulated in 2020 are expected to support household spending going forward

Household savings rose sharply in 2020, especially during periods at Alert Levels 3 and 4 when spending opportunities were limited (figure 2.6). Strong savings were aided by government policy measures, most notably the Wage Subsidy, which prevented a more significant decline in household incomes over 2020. While household saving has returned to around its usual level, there remains a large stock of savings accumulated in 2020 that has yet to be drawn upon, and this is expected to support household spending going forward.

Strong savings were aided by government policy measures, most notably the Wage Subsidy, which prevented a more significant decline in household incomes over 2020.

House price inflation has been higher than expected, but is assumed to moderate

Household spending has also been supported by rapid house price growth, which has been stronger than we expected. Investor demand appears to have fallen, but ongoing demand from owner-occupier buyers has supported continued price growth. Our central forecast is for house price inflation to ease over the coming year as prices adjust to reflect significantly lower population growth, higher mortgage interest rates, increased housing supply (figure 2.7), and the Government's housing policy measures.

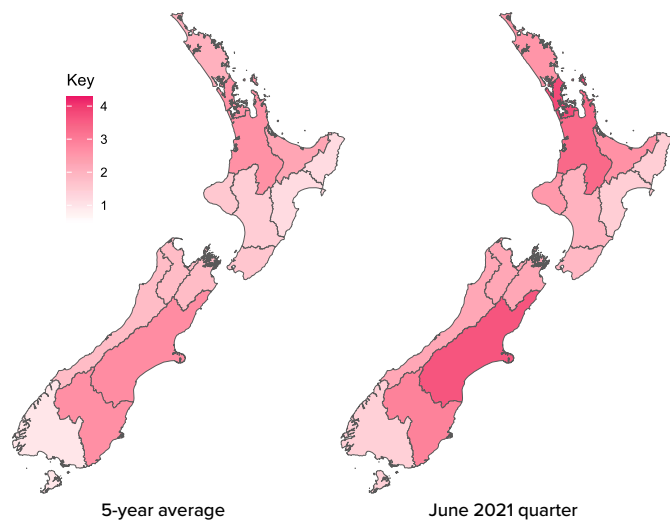
If the Reserve Bank were to further tighten loan-to-value ratio (LVR) restrictions, as announced for consultation in early August, this would also weigh on house price inflation. House prices are assumed to decline moderately in the latter part of the projection (figure 2.8). This reflects our assessment that house prices are unsustainable at their current levels, although the timing and magnitude of price declines are highly uncertain (discussed in Box A).

Global demand for our commodity exports is also supporting economic growth

The global recovery and continued demand for our commodity exports are also contributing to current economic momentum. The outlook for growth in many of our trading-partner economies has improved since the start of the year, notably due to optimism surrounding the vaccine rollout. Growth in China has also been strong, although it has moderated more recently. While prices have eased at recent Global Dairy Trade auctions, dairy prices remain high. Meat and log prices are also elevated. Strength in our export prices means that New Zealand's terms of trade remain high, despite the over 35 percent increase in oil prices since the start of the year (figure 2.9). The New Zealand dollar trade-weighted index (TWI) has remained near its level at the time of the May *Statement*, despite a widening in interest rate differentials. This has further supported exporter returns.

Figure 2.7

Residential building consent issuance (number per 1,000 working-age people, seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Note: By regional council area. For New Zealand as a whole, there were 3 consents issued for every 1,000 working-age people (aged 15 and over) during the June 2021 quarter. This compares to the 5-year average of 2.3.

Figure 2.8

House price inflation (annual)



Source: CoreLogic, RBNZ estimates.

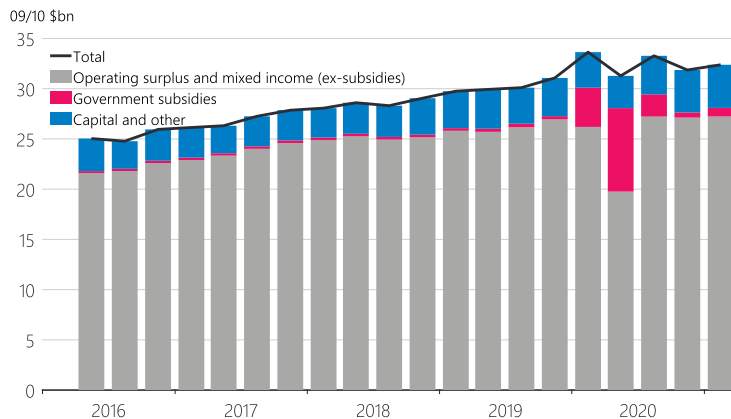
Figure 2.9

Dubai oil price



Source: Reuters.

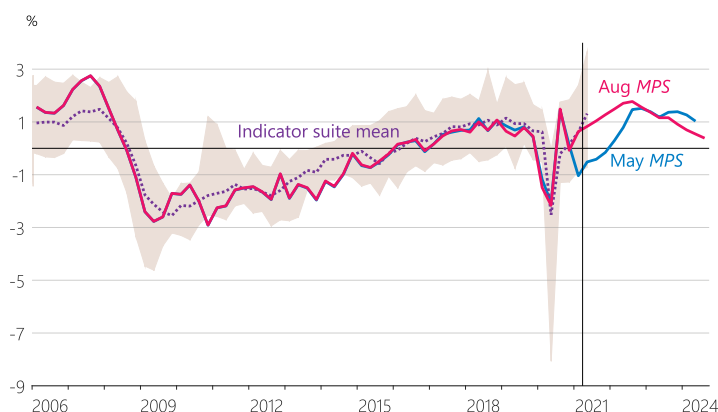
Figure 2.10
Quarterly non-financial business sector income
(indicative, nominal, seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

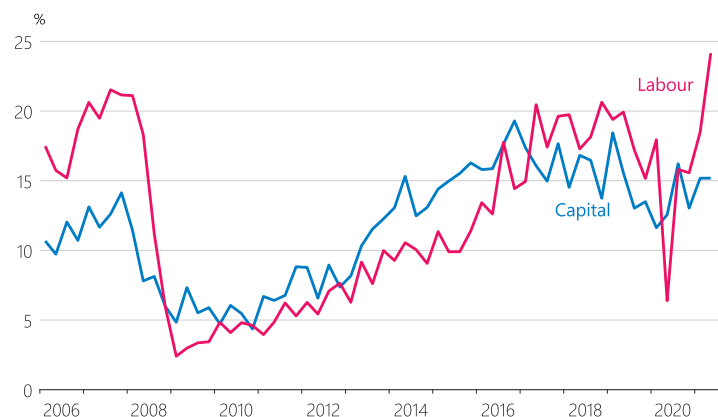
Note: The separation of 'government subsidies' from 'operating surplus and mixed income' in this chart is indicative only. It treats all government subsidies as having gone to non-financial businesses, when some would have been received by other sectors (e.g. the financial sector). This does not impact the total or the 'capital and other income' series.

Figure 2.11
Output gap and indicator suite
(share of potential)



Source: NZIER, Stats NZ, RBNZ estimates.

Figure 2.12
Factors limiting business expansion
(% of respondents, seasonally adjusted)



Source: NZIER, RBNZ estimates.

Note: Chart shows the share of businesses that reported labour or capital as the main factor limiting their ability to expand turnover, when responding to the Quarterly Survey of Business Opinion.

Business investment is recovering

Business investment recovered sharply in the March 2021 quarter, after being weak for much of 2020. As with household consumption, some of the recent increase likely reflected timing dynamics, such as delays due to supply-chain disruptions.

The recent increase in investment is consistent with strong export prices, reduced uncertainty, increasing capacity pressures, and improved profitability (as reflected in stronger-than-expected corporate income tax revenue). Nationwide business income has recovered to around pre-COVID-19 levels in nominal terms, although this will mask regional and industry divergences (figure 2.10). Business sector income actually increased during the quarters when New Zealand was at higher alert levels, once the impacts of the Wage Subsidy and other government support programmes are included. This explains some of the resilience of the business sector since the outset of the pandemic, and also why the adverse second-round effects from the periods at higher alert levels were not as severe as initially feared.

Pressure on available resources is increasing, particularly in the labour market

Business investment is expected to increase further and reinforce economic growth over the medium term. A key driver of increased business investment is further pressure on available resources, with our output gap indicator suite suggesting that the degree of spare capacity in the economy is much less than previously assumed (figure 2.11). Survey measures and discussions with businesses indicate pressures are particularly pronounced in the labour market, with businesses citing labour as the factor most limiting their production to a degree not seen since the 2000s (figure 2.12).

Employment is at or above its maximum sustainable level

Labour constraints partially reflect border restrictions and the difficulty in matching workers in New Zealand to available jobs, due to both regional and skill mismatches. Consistent with this, the unemployment rate has declined steadily since late 2020, whereas job vacancies remain high. Wage inflation has begun to increase, consistent with the degree of tightness in the domestic labour market (figure 2.13).

We now assess employment as being at or above its maximum sustainable level, as discussed in more detail in chapter 4. The employment rate is expected to increase gradually over the coming year, reflecting continued strength in economic activity (figure 2.14). It is then assumed to ease slightly over the second half of the projection, as capacity pressures normalise.

Inflation has increased due to temporary factors and rising capacity pressures

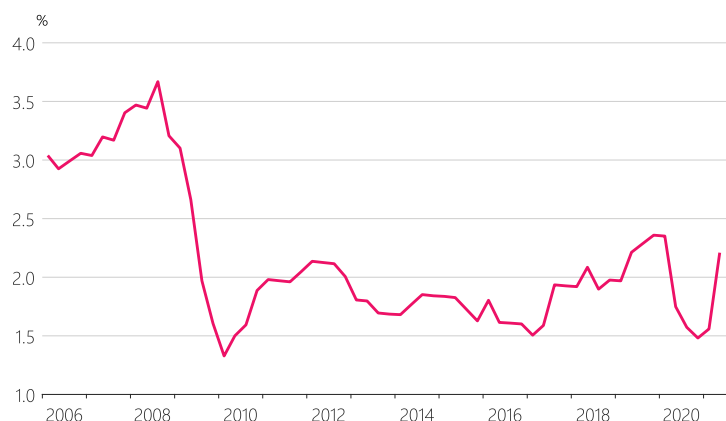
With stronger pressure on resources, high housing demand, supply-chain disruptions and sharp increases in oil prices, annual CPI inflation increased to 3.3 percent in the June 2021 quarter. While we had anticipated some increase in inflation, this increase was larger than expected.

Labour constraints partially reflect border restrictions and the difficulty in matching workers in New Zealand to available jobs – due to both regional and skill mismatches.

Figure 2.13

Annual wage inflation

(Labour Cost Index, private sector)



Source: Stats NZ.

Figure 2.14

Employment rate

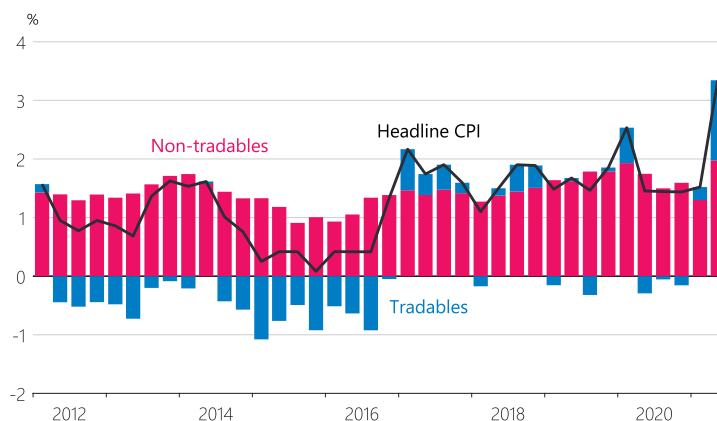
(share of working-age population, seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Figure 2.15

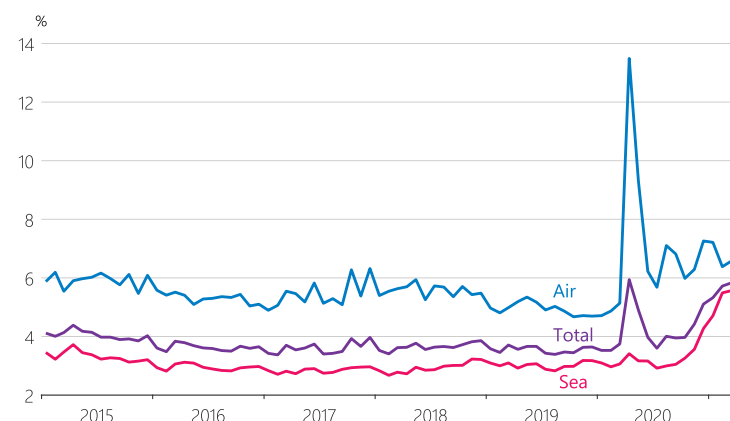
Contributions to annual CPI inflation



Source: Stats NZ, RBNZ estimates.

Figure 2.16

Shipping costs for consumer goods (share of import value)

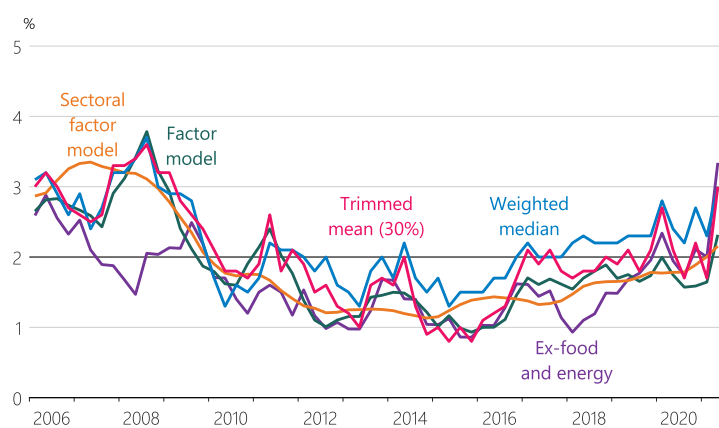


Source: Stats NZ, RBNZ estimates.

Note: These series are calculated by taking the value of consumer merchandise imports including freight and insurance costs (CIF) and subtracting the value reported for duty (VFD), which excludes these costs. It is expressed as a percentage of the VFD figure.

Figure 2.17

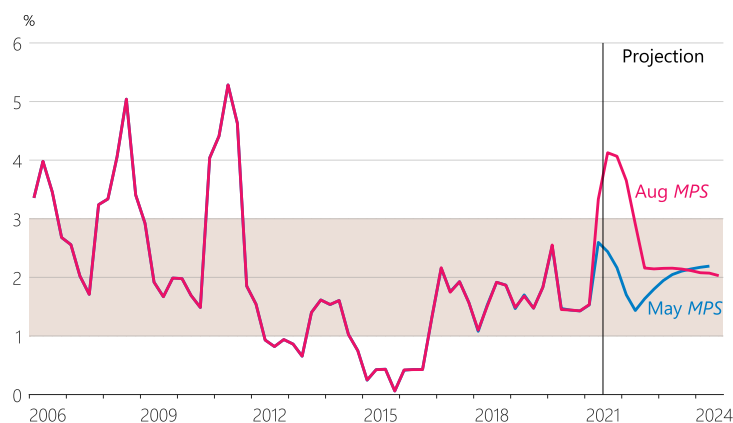
Core inflation measures (annual)



Source: Stats NZ, RBNZ estimates.

Figure 2.18

Annual CPI inflation



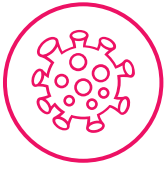
Source: Stats NZ, RBNZ estimates.

The increase in the June 2021 quarter was accounted for by both tradable and non-tradable components (figure 2.15). Supply-chain disruptions (as discussed in the May *Statement*) are continuing to exert upward pressure on prices via much higher shipping costs and difficulty sourcing some goods (figure 2.16). This is particularly notable in the construction sector where inflation is high.

Strong house price inflation has led to continued increases in the prices of housing-related components, including rents. The increase in petrol prices contributed 0.6 percentage points to the annual rate of inflation in the June quarter, reflecting the increase in global oil prices and the sensitivity of headline inflation to oil price movements. The latest increase in the minimum wage has also contributed to higher prices for some products. Looking through short-term volatility, measures of core inflation have increased after being below the mid-point of the target band for over a decade (figure 2.17).

CPI inflation is expected to increase temporarily in the near term, before easing gradually

CPI inflation is expected to increase further in the near term, due to continued momentum in the factors already discussed – capacity pressures, housing-related components, supply-chain disruptions – and additional increases in specific factors, such as council rates, electricity and petrol prices. Inflation is expected to moderate towards 2 percent over the next three years, as temporary factors unwind and capacity pressures ease (figure 2.18). However, there are risks to this outlook, as discussed in the following section.



Key risks to the economic outlook

The appropriate monetary policy stance depends not only on our central outlook for the economy, but also on our assessment of the balance of risks to this outlook. The risks discussed below are not intended to be exhaustive, but instead illustrate some factors that could substantially alter the economic and monetary policy outlook.

COVID-19 resurgence in New Zealand

The largest downside risk to the New Zealand economy remains a health risk. New Zealand managed to eliminate community cases of COVID-19, which enabled a return to many of our usual activities and ways of working. A widespread outbreak of COVID-19 may put our economic recovery at risk, especially if our vaccination levels are insufficient.

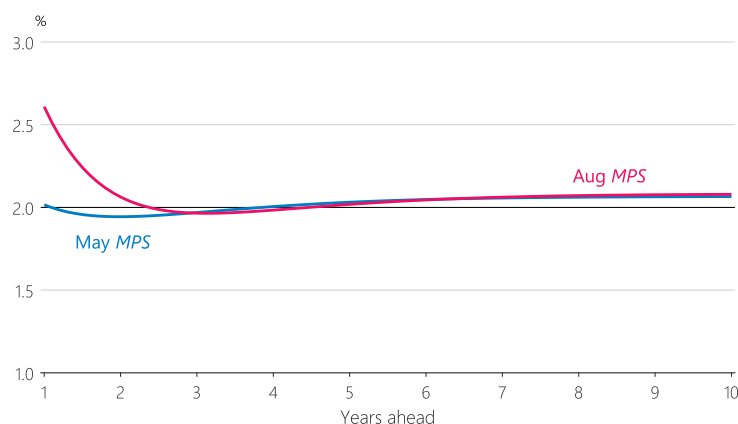
However, we have learned from past experience that the economy is resilient to periods at higher alert levels if there is significant government support provided, and alert levels can be lowered relatively quickly. We have seen the effectiveness of the Wage Subsidy in keeping people employed and preventing significant adverse second-round effects (e.g. from periods of prolonged high unemployment, and business closures), and the ability of businesses to adapt their operations to meet social distancing requirements. These factors lessen the likely impact of COVID-19 outbreaks on our inflation and employment objectives.

Higher inflation becomes embedded

We currently assume supply-chain disruptions start to ease during 2022. There is a risk that these disruptions will last longer, and continue to exert upward pressure on headline inflation. There is also a risk that high inflation outcomes will lead to an increase in expected inflation, and become embedded in wage- and price-setting behaviour to a greater extent than currently assumed.

While inflation expectations have shifted a lot at shorter horizons (expectations tend to reflect recent outcomes at these horizons), expectations remain anchored over the medium and long term (figure 2.19).

Figure 2.19
Inflation expectations
(annual)



Source: RBNZ estimates.

Note: See 'Inflation expectations and the conduct of monetary policy in New Zealand' (Lewis, McDermott and Richardson, 2016) for a description of the inflation expectations curve.

Economic effects of easing border restrictions

The economic effects of easing border restrictions are uncertain. In recent quarters we have seen capacity pressures return, even with our borders largely closed to international visitors. This is partly due to the onshoring of travel spending by New Zealanders. This creates additional uncertainty about economic activity once border restrictions ease, as we will see more international visitors spending here but also more New Zealanders spending abroad.

Limited access to global labour has also contributed to labour market tightness. While easing border restrictions will likely alleviate specific labour shortages in some sectors, it is unclear whether it will result in a net flow of people into New Zealand.

Uncertainty regarding the longer-term economic implications of COVID-19

The emergence of highly contagious variants of COVID-19 reduces the likelihood that New Zealand will be able to ease border restrictions in the future and be COVID-19 free. This will lead to ongoing and challenging trade-offs between public health, economic outcomes, and mobility.

New Zealand's economy has not experienced a sustained period of community transmission of COVID-19 in the absence of significant mobility restrictions. Prevalence of the disease may result in permanent changes in the way we work and interact, which could lower the productive capacity of the economy on an enduring basis. How the global economy adapts to the ongoing presence of COVID-19 will also have significant implications for the New Zealand economy, via trade, financial, and confidence channels.



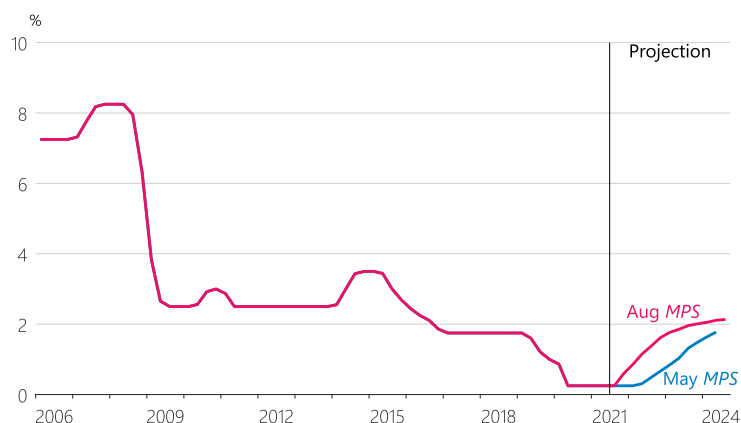
Monetary policy outlook

Monetary policy has been playing a supporting role in the economic recovery, with fiscal policy taking the lead in bridging households and businesses through periods at higher alert levels. Given the current and expected strength in the labour market and increase in inflationary pressures, our central projection implies that monetary stimulus should be reduced in order for the MPC to meet its economic objectives. If economic conditions evolve as assumed in our projection (see chapter 5), we anticipate that the OCR will eventually need to be raised to a level where it is neither stimulatory nor contractionary (figure 2.20). Chapter 4 discusses our assessment of the level of this ‘neutral’ interest rate.

The decision at the July *Review* to halt purchases of government bonds under the large scale asset purchase (LSAP) programme implies no more additional stimulus is being provided by the flow of purchases. However, the stock of assets purchased under the programme is assumed to continue to provide a degree of stimulus (see chapter 3).

Figure 2.20

Official Cash Rate



Source: RBNZ estimates.



Box A

House price sustainability

House prices in New Zealand have increased rapidly over the past year from an already elevated level. A confluence of factors suggests that house prices are above what is sustainable. While house price inflation has slowed in the past three months, momentum has been stronger than expected. A moderation in house prices is expected over coming years, but the precise timing of this is uncertain.

House prices appear to be above their sustainable level

- Underlying demand for housing due to population growth has declined significantly since the outbreak of COVID-19 last year. While New Zealand citizens who returned from abroad before the pandemic stayed on and others returned early in the outbreak, this inflow was short-lived. Border restrictions have since limited inward migration, and there has been a small but steady flow of departing residents. Net migration is not anticipated to return to pre-COVID-19 levels over the next few years, even as border restrictions are eased.

- Meanwhile, house building is at record high levels (figure A1). Residential building consents and discussions with construction sector businesses suggest there is a significant pipeline of new housing supply coming. Previous large increases in housing supply in New Zealand, such as that during the 1970s, reduced real house prices. In recent decades, several other countries have experienced declining house prices following significant increases in supply.

Building consents data suggest that by the middle of next year, the total number of houses will be growing at its fastest pace since data

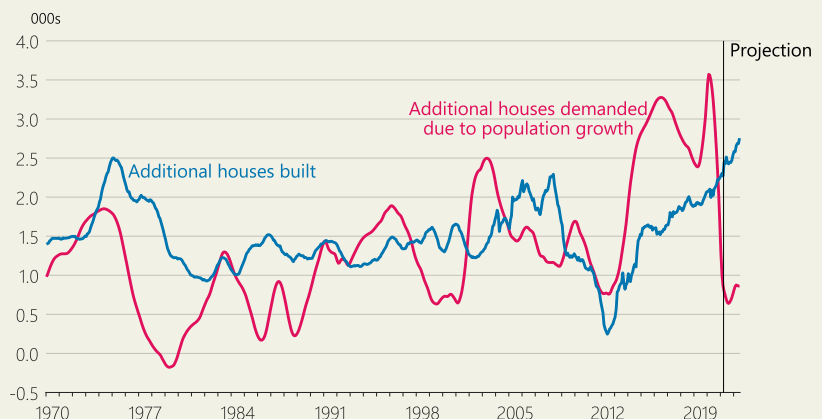
became available in the early 1960s. Policy changes that significantly ease land-use restrictions will encourage continued strong levels of building.¹ Previous relaxations of land-use restrictions have contributed to sustained increases in new supply.²

Housing supply has not kept up with population growth over most of the past decade, increasing house prices and necessitating larger households on average. We consider this undersupply to already be reflected in current house prices. As a result, further strong house building will put downward pressure on house prices, even given the historical undersupply.

Figure A1

New supply of and demand for houses

(indicative, monthly, 12-month moving average)



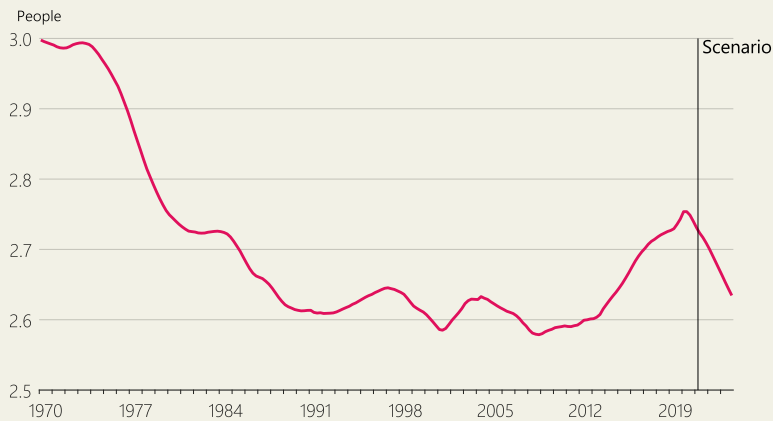
Source: Stats NZ, Electricity Authority, RBNZ estimates.

Note: 'Additional houses built' is based on the change in electricity connections to residential properties, while residential building consents are scaled to produce estimates prior to 2003 and a projection for the next year. 'Additional houses demanded' is population growth divided by people per dwelling (see figure A2).

1 For example, the **National Policy Statement on Urban Development** directs local authorities to enable greater supply and ensure that planning is responsive to changes in demand. It also requires councils to remove overly restrictive rules that affect urban development.

2 For example, the Auckland Unitary Plan unlocked significant development potential within Auckland. Policy changes following the 2010 and 2011 Canterbury earthquakes allowed a rapid recovery in Canterbury's housing stock. They also enabled more responsive supply, which appears to have suppressed house price growth in Canterbury over the past decade.

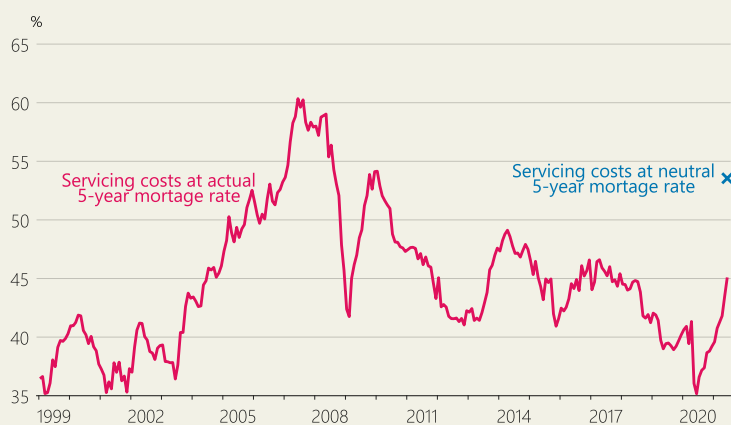
Figure A2
Outlook for people per dwelling



Source: Stats NZ, Electricity Authority, RBNZ estimates.

Note: The people per dwelling ratio is the population over an estimate of the number of dwellings based on electricity connections to residential properties. As in figure A1, residential building consents are scaled to produce estimates prior to 2003 and forecasts for the coming year.

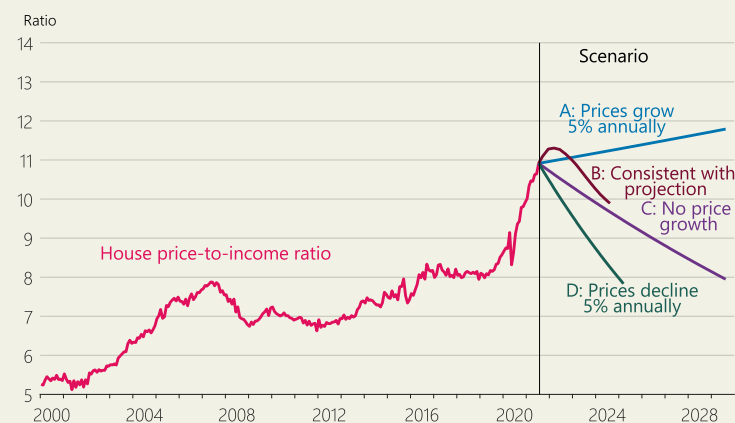
Figure A3
Estimated debt servicing cost for new buyers as a share of income



Source: RBNZ estimates.

Note: Debt servicing costs include both interest and principal repayments, based on a 30-year mortgage term. Estimates are for buyers purchasing at the median selling price with a 20 percent deposit.

Figure A4
Median house price-to-income scenarios



Source: Stats NZ, REINZ, RBNZ estimates.

Note: All scenarios assume that median household disposable income grows at 4 percent annually, which is close to the 10-year average annual growth rate. Scenario B assumes that the median house price grows at the same rate as our projection for the CoreLogic House Price Index. The line for Scenario B ends in the September 2024 quarter, at the end of our projection horizon.

Consistent with an easing of housing market imbalances, household sizes are likely to decline (figure A2). As a hypothetical exercise, we estimate that if population growth and the construction of new dwellings continue at their current rates over the next few years, people per dwelling would return to its 2014 level by the end of 2023.

- Meanwhile, a range of other factors are likely to moderate demand. The return from investment in housing will be reduced significantly by the removal of the tax deductibility of interest from rental income and the extension of the brightline test. Along with tighter LVR requirements, these policy changes have reduced investor buying activity in the housing market. The share of new mortgage lending for investment property has declined to 17 percent, from 26 percent at the beginning of 2021.
- Interest rates are low relative to their neutral levels, and therefore below where they will eventually return (see chapter 4). Against the backdrop of a long-run downward trend in interest rates, monetary policy was eased at the outset of the COVID-19 pandemic to support New Zealand's economy. As interest rates rise in response to the improving economic outlook, this will dampen demand for housing in several ways.

Higher interest rates will make investing in housing relatively less attractive than interest-bearing assets. Higher interest rates will also constrain the ability of new buyers to service debt while leaving enough income for consumption. Households are borrowing more relative to their incomes, in part because at the current low interest rates debt servicing costs are low for new buyers. However, if interest rates increase back to a more neutral level, borrowing capacity would be reduced (figure A3). In this scenario, demand to purchase houses at current prices would fall.

We anticipate a realignment in house prices

With house prices above what is sustainable, some form of realignment is anticipated. However, as seen recently, momentum in house price growth can persist even when prices look disconnected from the fundamental factors that should determine them. This reflects that sentiment, expectations, and prevailing narratives surrounding the housing market can have a significant bearing on housing demand and house prices. The further house prices rise above their sustainable level, the larger the required realignment will need to be.

It is not clear when and how a realignment of house prices will occur. Growth in household incomes could lift the sustainable level over time to a point where current prices would be sustainable. However, even if house prices stay at their current levels and incomes grow as they have historically, it would take eight years for house prices to return to the same level relative to incomes as in early 2020 (figure A4). Alternatively, falls in house prices could facilitate a faster adjustment towards a more sustainable level.

We expect house price inflation to moderate significantly over the coming quarters, consistent with the factors discussed above: low population growth, high levels of house building, higher interest rates, and tax policy changes. In our projection, house prices are assumed to begin to fall modestly from late 2022 (see figure 2.8). A more significant fall in prices is possible, but at the same time, momentum in the market could prove more resilient than we expect.

Monetary policy is focused on employment and inflation objectives

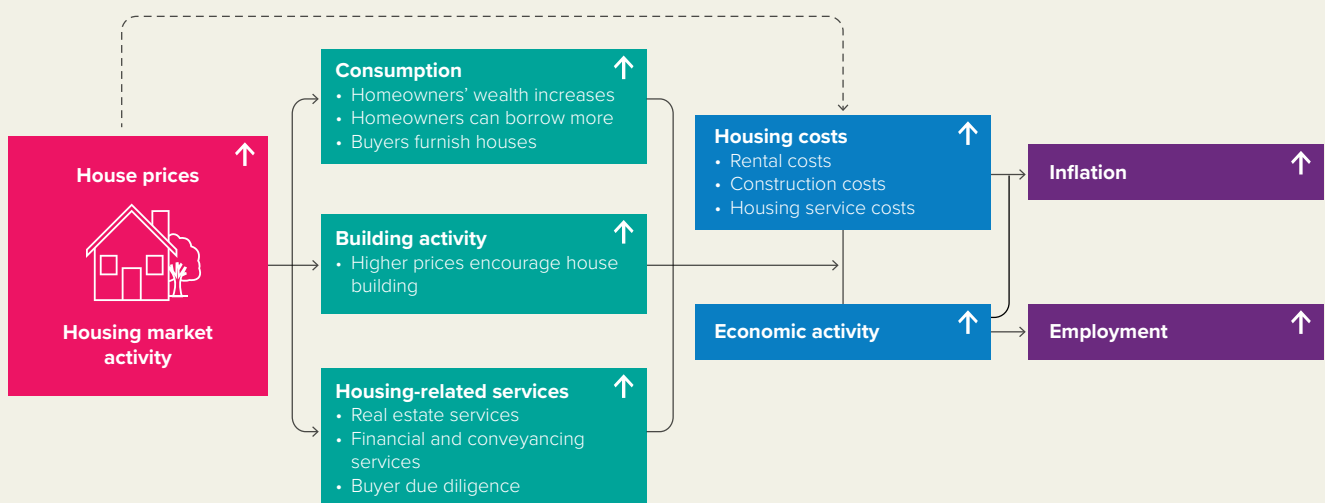
The MPC sets monetary policy to achieve its inflation and employment objectives in the *Remit*. It considers the outlook for the housing market because house prices can influence broader economic activity, employment, and consumer price inflation (figure A5). Currently, even if there is a material slow-down in house price inflation, we anticipate that monetary policy stimulus will need to be gradually reduced in order to meet the *Remit* objectives. This reflects broader momentum and inflationary pressures in the economy (see chapter 2).

Housing market outcomes can also have a bearing on financial stability, which the MPC must have regard to when setting monetary policy. However, the MPC considers that macroprudential policy should be the primary instrument for addressing the risks from household indebtedness (see chapter 3 of the November 2019 *Statement*). The Reserve Bank **will be consulting** on a proposal to tighten restrictions on mortgage lending at high LVRs. It also intends to consult on introducing debt-to-income (DTI) restrictions and/or floors on the interest rates banks use in their mortgage serviceability assessments. These measures are intended to ensure borrowers are resilient to a range of future economic and financial conditions.

The MPC remains focused on its inflation and employment objectives. The MPC will continue to assess the impact of monetary policy on sustainable house prices, as required in the *Remit*.

Figure A5

Flow-on impacts from house prices to the broader economy



An aerial photograph showing a wide, shallow river with light-colored, rippling water. The river flows from the top left towards the bottom center. To the right of the river is a dense, lush green forest. The left bank of the river is composed of dark, pebbly or sandy material. The overall scene is captured from a high angle, looking down on the landscape.

Monetary
conditions

CHAPTER
03

CHAPTER 3

Monetary conditions

During 2020, the MPC cut the OCR to a record low level of 0.25 percent, and introduced additional monetary policy (AMP) tools to support New Zealand's economy through the COVID-19 pandemic (table 3.1). These actions were summarised in the

February 2021 Statement, and have influenced economic activity through several financial market channels that were summarised in the **May 2021 Statement**. A forthcoming *Bulletin* article will go into further detail on the full suite of available monetary policy tools.

Table 3.1:

Monetary policy tools used during the COVID-19 pandemic

Tool	Description	Date implemented
OCR	The OCR was lowered from 1.00% to 0.25%.	16 March 2020
Forward guidance	A commitment was made that the OCR would not change for at least 12 months from 16 March 2020.	16 March 2020
Large Scale Asset Purchase (LSAP) programme	The Reserve Bank undertook to purchase a significant amount of New Zealand government and Local Government Funding Agency bonds, in order to lower long-term interest rates.	23 March 2020 (\$30b limit) 7 April 2020 (\$33b limit) 13 May 2020 (\$60b limit) 12 August 2020 (\$100b limit)
Funding for Lending Programme (FLP)	The FLP is a low-cost funding facility for banks. It is equivalent in size to 6% of participants' loans. The facility lowers banks' funding costs directly, and also indirectly by reducing the cost of their other sources of funding.	7 December 2020

The MPC has determined that less stimulus is required

Monetary policy remains accommodative (see chapter 2). However, the MPC has recently reduced the level of monetary stimulus. At the July 2021 *Review*, the MPC halted further purchases of government bonds under the LSAP programme. This reduced monetary stimulus, primarily by bringing forward market participants' expectations of future OCR increases. Financial market participants had expected purchases to continue for a while longer, and for the OCR to increase after purchases had ceased.

Our economic projections imply that increases in the OCR over the next year will be necessary for the MPC to meet its inflation and employment objectives.

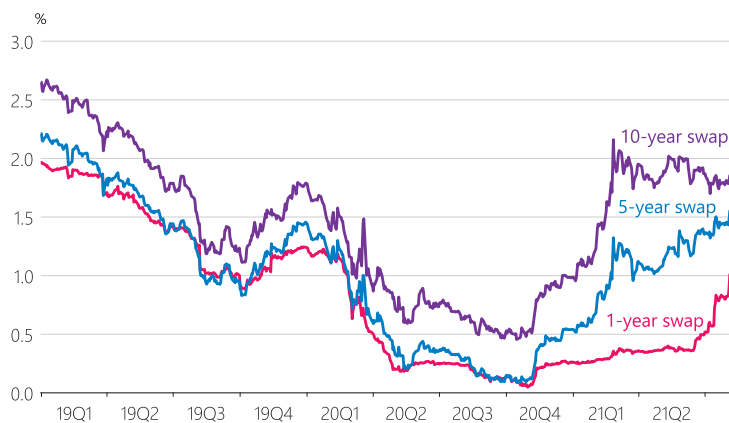
Financial markets have responded to strong economic data

Financial conditions have been tightening for several months due to expectations of OCR increases. Stronger economic data and the accelerating vaccine rollout globally have increased confidence that economic conditions will eventually warrant higher interest rates. Accordingly, longer-term wholesale interest rates have increased since the start of the year, as evidenced by increases in the 5-year and 10-year swap rates (figure 3.1).

Shorter-term rates remained anchored during the first half of this year, but have increased during the past two months on the release of domestic GDP, inflation, and employment data, which have shown that inflationary pressures and employment were higher than previously anticipated. This has led financial market participants to anticipate an earlier increase in the OCR than was expected only a few months earlier. The 1-year swap rate rose by more than 70 basis points in a two-month period.

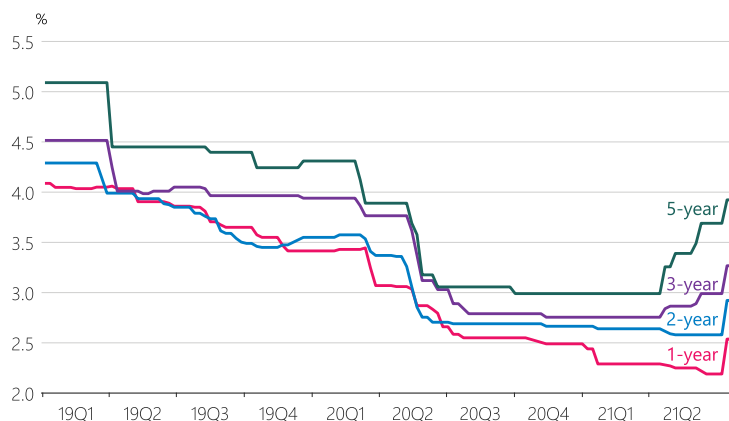
Interest rates paid by businesses and mortgage borrowers, as well as those paid to depositors, have begun to increase in response to rising wholesale interest rates. Nevertheless, bank interest rates remain low. For example, mortgage interest rates remain below their pre-pandemic levels (figure 3.2).

Figure 3.1
Swap rates



Source: Bloomberg.

Figure 3.2
Mortgage rates



Source: interest.co.nz.

Note: The rate shown for each term is the average of the latest 'special' rates on offer from ANZ, ASB, BNZ, and Westpac.

The effects of additional monetary policy tools will persist

The Reserve Bank currently holds \$54 billion of bonds purchased under the LSAP programme. Although the Reserve Bank has stopped purchasing bonds, the stock of bonds held on the Reserve Bank's balance sheet, and the associated balances in the settlement cash system, will continue to provide stimulus.

Furthermore, the FLP remains in place, with the initial allocation available until June 2022. The additional allocation, which applies to banks that grow their lending, is able to be drawn upon until December 2022. The interest rate on funds borrowed under the FLP is set at the OCR, so if the OCR changes, so does the cost of funding via the FLP.

The term of FLP loans is three years, which means that some of the low-cost funding provided by the FLP could be on banks' balance sheets until 2025, directly reducing banks' funding costs. However, the direct effect of the FLP is expected to be relatively small. The more significant, indirect effect of the FLP – due to the FLP providing a lower-cost alternative to other funding and therefore pushing down its cost – is expected to persist only until the end of the periods during which the FLP can be drawn upon.

The FLP's continued downward pressure on bank funding costs is demonstrated in term deposit rates, which have remained low relative to comparable wholesale rates (figure 3.3).

An outcome of the ongoing stimulus provided by the LSAP programme and the FLP is that the OCR will need to be higher than otherwise to achieve a given overall level of stimulus.

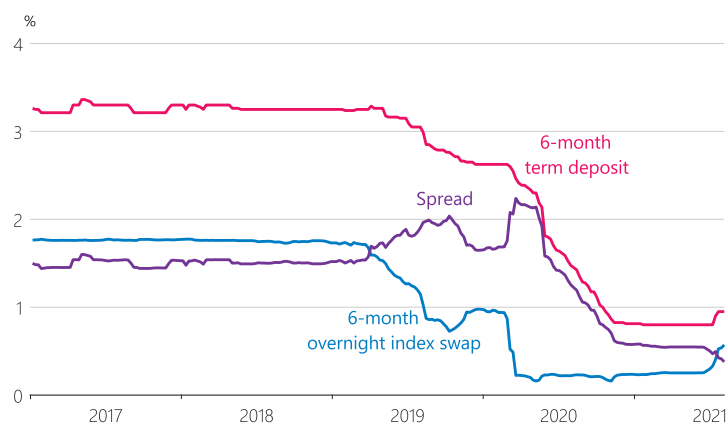
The OCR is the preferred tool for adjusting monetary policy

The LSAP programme and the FLP are geared towards providing stimulus for a sustained period of time. In contrast, the OCR is a much more flexible tool, which can be adjusted more frequently and calibrated more precisely according to the level of monetary stimulus required. The OCR is currently the preferred tool for adjusting monetary policy.

When setting monetary policy, the MPC will continue to take into account how broader interest rates are evolving. Given monetary policy has tightened, the Reserve Bank will consider how previous periods of monetary policy tightening have affected the economy, and how the economy has changed since then. For example, transmission of tighter monetary policy to the household sector is likely to be influenced by changes in household balance sheets and borrowing behaviours (see box B).

Figure 3.3

6-month term deposit spread to 6-month OIS



Source: interest.co.nz, Bloomberg.



Box B

Transmission of tighter monetary policy through the household cash flow channel

One example of how higher interest rates will affect the economy is through repricing of mortgages. The transmission of monetary policy depends to an extent on the state of household balance sheets at the time, as this will influence the level of household consumption – a key source of demand in the economy. This box compares the current environment to previous periods of rising interest rates.

The 1-year mortgage rate has so far increased by around 35 basis points from its recent lows. If the OCR returns to the Reserve Bank's current estimate of its 'neutral rate' (see chapter 4) the 1-year mortgage rate could increase materially. Table B1 gives an indication of the increases in mortgage payments that may be faced by recent borrowers under different interest rate scenarios.

OCR increases are passed through to mortgaged households' interest payments more quickly when borrowers are on shorter fixed terms. Compared to the beginning of the two most recent monetary policy tightening cycles (discussed below), the distribution of mortgages' time to reprice is broadly similar, with around 78 percent due to reprice within one year. However, the level of debt is higher relative to disposable income; the quantity of debt due to reprice within one year has risen to 120 percent of household income (figure B1).

Table B1

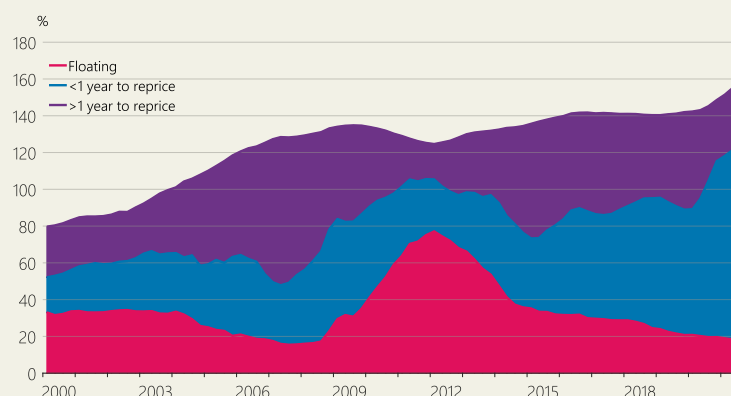
Effects of interest rate changes on mortgage payments

(principal and interest for a \$700,000 mortgage on a 30-year term)

Interest rate (%)	Weekly payments (\$)	Increase in payments (\$)	Annual payments (\$)	Increase in payments (\$)	Mortgage payments % of take-home pay*
2.55 (Current 1-year fixed rate)	642	-	33,391	-	36.1%
3.00	681	+39	35,394	+2,003	38.3%
4.00	771	+129	40,076	+6,686	43.3%

Note: \$700,000 is a representative loan amount for a recent first-home buyer. We have assumed the borrower has a debt-to-income (DTI) ratio of 5, consistent with the average for first-home buyers over the past 12 months. Calculations in the right-hand column therefore assume a dual-income household with gross income of \$140,000, student loan deductions at the standard rate, and 4% KiwiSaver contributions.

Figure B1

Mortgage debt by time to reprice*(% of household disposable income)*

Source: RBNZ.

The high starting point for debt means that a given increase in the OCR will have larger effects on mortgaged households' discretionary incomes than in previous tightening cycles. However, the quantity of household deposits is also higher, leading to a greater potential increase in interest incomes for those households with deposit savings.

By comparison, at the beginning of 2004, the quantity of debt due to reprice within one year was 65 percent of household income. The OCR increased from 5.00 percent to 8.25 percent between January 2004 and July 2007. The 1-year mortgage rate increased earlier on anticipation of OCR increases; it increased from 6.20 percent in July 2003 to 9.95 percent in March 2008. During this period, expectations that the OCR was near its peak began to build, causing longer-term mortgage rates to become lower than shorter-term mortgage rates. This lasted for about two years, and saw a large portion of borrowers fix at longer terms as interest rates rose.

The only significant period of OCR increases in the past decade began in March 2014. Banks had begun to increase their fixed mortgage rates six months prior to the first OCR hike, consistent with OCR expectations and the Reserve Bank's communications at the time. At the beginning of 2014, the quantity of mortgage debt due to reprice within one year was 97 percent of household disposable income. During this relatively short tightening cycle, which was subsequently reversed, the 1-year mortgage rate increased by around 80 basis points before declining again.

The background is an abstract composition of layered, textured surfaces. The upper portion is a warm, golden-brown color with fine, horizontal, wavy lines, resembling a close-up of a rock face or a piece of aged parchment. The lower portion features a more complex, darker brown and teal-green texture, with irregular, rounded shapes that look like water droplets or small pools of liquid. The overall effect is one of depth and organic complexity.

Special topics

CHAPTER
04

CHAPTER 4

Special topics

Prior to each *Statement*, the MPC is provided with analysis of some topical issues.

Topics for the August *Statement* included:

1. the recovery of New Zealand's labour market; and
2. the neutral interest rate.

1

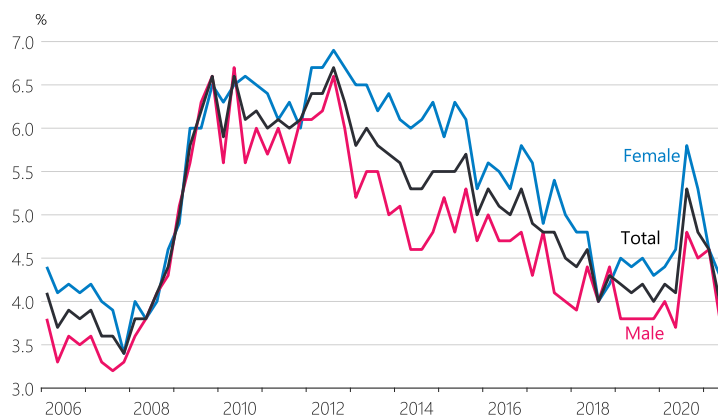
The recovery of New Zealand's labour market

The labour market has fared better than expected since the outset of the COVID-19 pandemic. Supported by significant fiscal and monetary stimulus, businesses have been able to retain staff and stay solvent through periods at higher alert level restrictions. The strong rebound in economic activity, and border restrictions that have limited access to workers from abroad, have led to a tighter labour market than prior to the pandemic.

The unemployment rate was 4.0 percent in the June 2021 quarter, after peaking at 5.3 percent in the September 2020 quarter (figure 4.1). Across sex and ethnicities, except Pacific peoples, unemployment rates have returned to around pre-COVID-19 levels.

When considering a broad range of labour market indicators, employment is assessed as being at or above its maximum sustainable level. This assessment takes into account that the level of maximum sustainable employment (MSE) is likely lower in the current environment than it would be if border restrictions and limited labour mobility were not impinging on the economy.

Figure 4.1
Unemployment rate by sex
(seasonally adjusted)



Source: Stats NZ.

The recovery started off uneven and has been driven by compositional changes

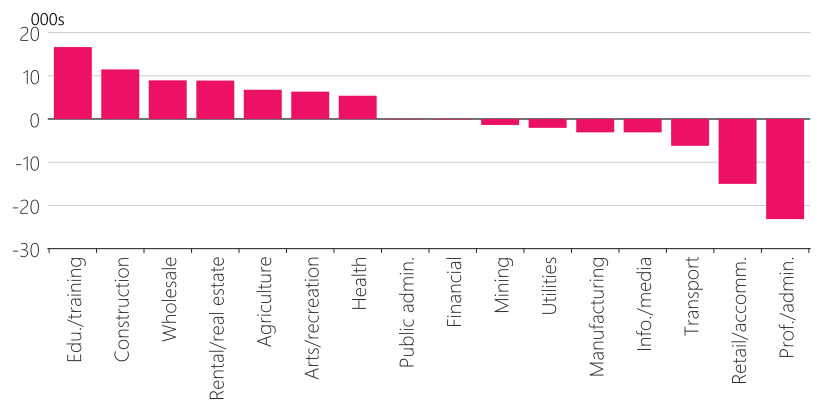
The COVID-19 pandemic and public health response measures have hit tourism-related industries particularly hard. As a result, employment in most of these industries is still below its pre-COVID-19 levels. This is particularly the case in retail trade, accommodation and transport, as well as professional and administrative services, which include travel agents and tour operators (figure 4.2). The unemployment rates in most regions with a greater reliance on tourism are also above their pre-COVID-19 levels (figure 4.3).

The total number of employed people mainly comprises paid employees, self-employed, and working employers. Paid employment has been the most affected by the COVID-19 pandemic. However, the negative impact from the decline in paid employment has been partially offset by an increase in self-employment (figure 4.4), particularly for women. Flexibility between types of employment helped limit the increase in the unemployment rate over 2020. Since its initial increase in the June 2020 quarter, self-employment has remained elevated. This has been the case even as paid employment opportunities have recovered across many regions and industries, with the number of job vacancies at record highs. This suggests that switching to self-employment has not just been a short-term response, but a more deliberate choice by workers.

Figure 4.2

Employment by industry

(cumulative change since 2020Q1, seasonally adjusted)



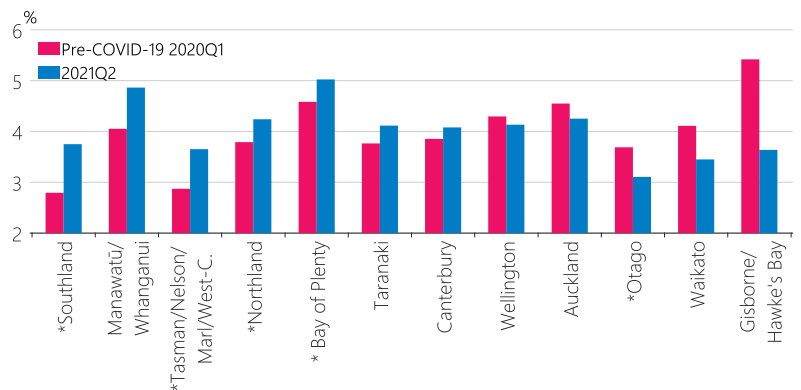
Source: Stats NZ, RBNZ estimates.

Note: The category 'not specified' is excluded.

Figure 4.3

Unemployment rate by region

(seasonally adjusted)



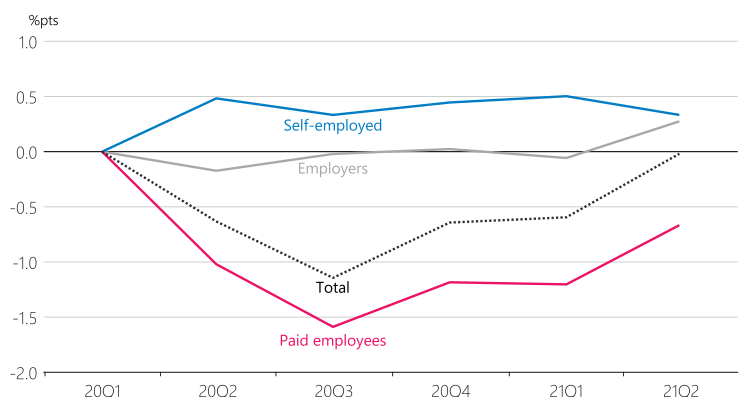
Source: Stats NZ, RBNZ estimates.

Note: Regions marked with an asterisk (*) had tourism expenditure in excess of 10 percent of regional GDP in the year to March 2020. Regions are sorted by the difference between the red and the blue bars (current minus pre-COVID-19 unemployment rate), with regions where unemployment is above pre-COVID-19 levels being on the left.

Figure 4.4

Employment rate by type

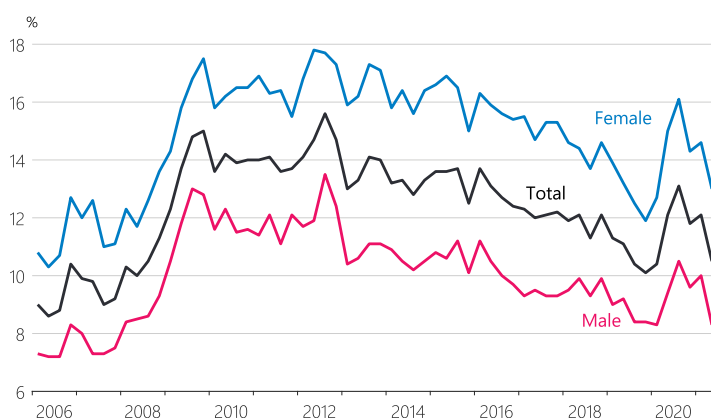
(cumulative change since 2020Q1, seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Some businesses that faced lower demand reduced employees' hours, partly by switching towards part-time from full-time roles.³ This is a typical response to a downturn in New Zealand, with businesses trying to avoid redundancies by reducing employees' work hours in the first instance.⁴ It has helped to keep people in jobs, but initially resulted in a significant increase in underemployment (the number of part-time employees who want to work more hours). However, recent strong labour demand has helped lower underemployment towards pre-COVID-19 levels, as the labour market recovery has become more broad based. The underutilisation rate – which includes unemployed and underemployed people – has nearly returned to pre-COVID-19 levels for both women and men (figure 4.5).

Figure 4.5
Underutilisation rate by sex
(seasonally adjusted)



Source: Stats NZ.

Employment is at or above its maximum sustainable level

Unemployment and underutilisation have recovered quickly and labour demand is significantly above pre-COVID-19 levels. The number of businesses intending to increase their workforces is high, and for many businesses insufficient labour is reported as being the main factor limiting their ability to increase output. Consistent with this, more businesses than ever are reporting that it is difficult to find skilled and unskilled labour. With the exception of hospitality, which continues to face reduced tourism demand, job vacancies are at or above pre-COVID-19 levels across all regions and industries.

While labour demand has been strong, employers have not been able to draw all the workers needed from the pool of unemployed or underemployed, or from outside the labour force. This reflects a combination of factors:

- Border restrictions have made it difficult for businesses to source the labour needed from abroad.
- Aggregate unemployment and underemployment have fallen significantly, but some pockets of weakness remain, in tourism in particular. Consequently, additional labour capacity within New Zealand seems to be concentrated mainly in workers who formerly worked in tourism-related industries or who are still working in those industries but for fewer hours. These people may not always have the skills needed for the jobs available, or may not be willing or easily able to move between regions for work.
- The increase in self-employed workers has reduced the pool of labour available for businesses needing to fill paid positions on a permanent basis.

³ In particular, accommodation and food services as well as rental, hiring, and real estate services (which include car rental) show higher ratios of part-time to full-time hours worked compared to pre-COVID-19 levels.

⁴ See Culling and Robinson (2020), 'Employment and hours worked adjustment in New Zealand's labour market', *Analytical Note*, Reserve Bank of New Zealand.

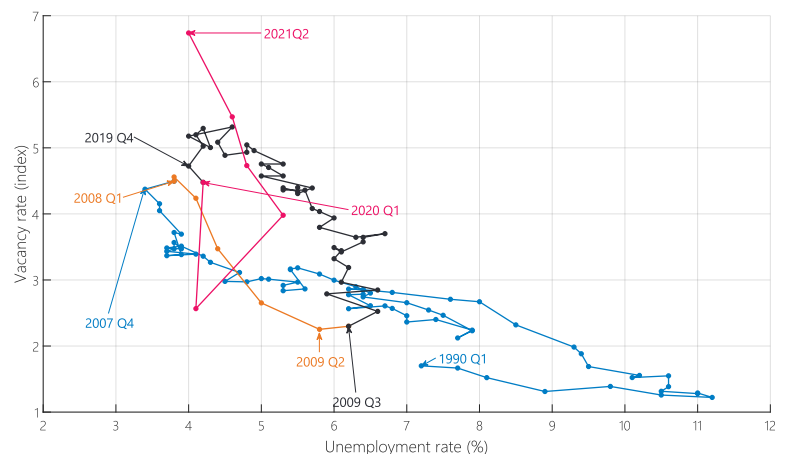
These factors suggest that the level of MSE is likely lower in the current environment than it would be if these constraints were not present. In other words, the labour market has been unevenly impacted by COVID-19 and, while these structural issues remain, increasing labour demand (such as through monetary policy stimulus) is likely to be less effective at lifting employment.

The flipside of a lower level of MSE is higher structural unemployment. The Beveridge curve is one concept that can show changes in structural unemployment (figure 4.6).⁵ For a given level of structural unemployment, a higher vacancy rate is associated with a lower unemployment rate and vice versa. The vacancy and unemployment rates move along a specific Beveridge curve. However, a Beveridge curve that has changed in steepness or position may suggest a different level of structural unemployment. Over recent quarters the vacancy rate has increased significantly, but declines in the unemployment rate have not kept pace, despite the sharp fall in the June 2021 quarter (the pink line in figure 4.6). This steepening of the Beveridge curve indicates that those who are not currently employed are not well matched to what employers are looking for in terms of skills and locations. This indicates higher structural unemployment and a lower level of MSE.

A lower level of MSE is also consistent with increased wage pressures at any given unemployment rate. Since the unemployment and underutilisation rates have fallen, the pool of available labour is even more limited and businesses will have to bid up wages to attract workers. The latest labour market data provide early evidence that this is beginning to occur. Annual wage inflation has increased to 2.2 percent, above its post-2010 average of 1.8 percent, and we expect further wage inflation over the coming quarters (see chapter 5).

Given that many labour market indicators are around their pre-COVID-19 levels and the level of MSE is likely lower than it was then, employment is currently assessed as being at or above its maximum sustainable level (figure 4.7).

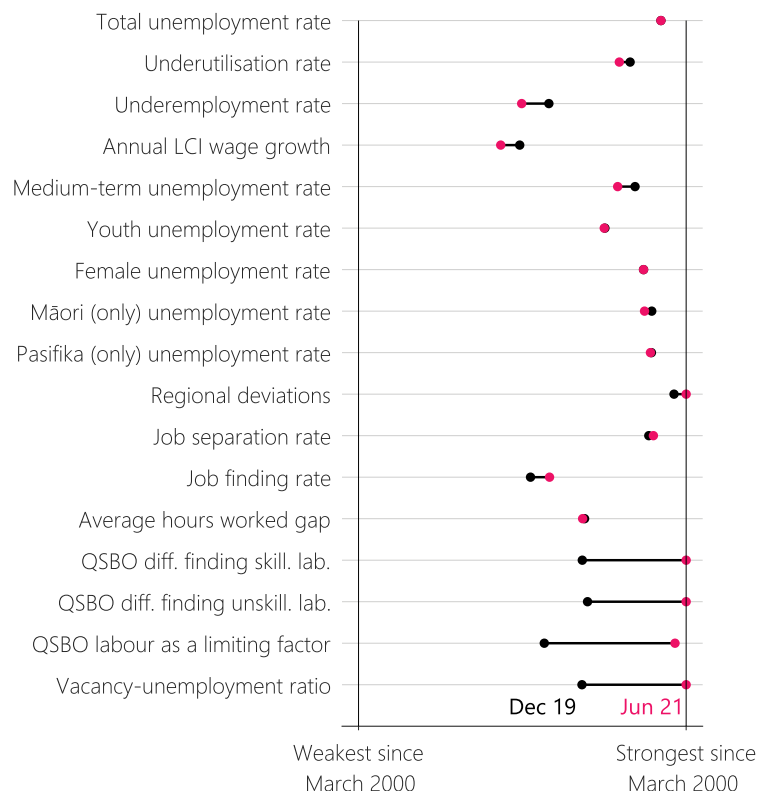
Figure 4.6
Job vacancies and unemployment (Beveridge curve)
(seasonally adjusted)



Source: Stats NZ, Ministry of Business, Innovation and Employment (MBIE), RBNZ estimates.

Note: The vacancy rate is defined as the number of job vacancies as a share of the labour force. Colours reflect different periods in recent history (pink: COVID-19; black: post-GFC; orange: GFC; blue: pre-GFC).

Figure 4.7
MSE indicators



Source: Stats NZ, MBIE, ANZ, NZIER, RBNZ estimates.

Note: The Reserve Bank uses a range of indicators when assessing MSE and regards the December 2019 quarter as a period when this selection of indicators pointed to employment being at MSE. However, current outturns should not be directly compared to 2019Q4 as the level of MSE is likely lower. The vertical lines on the left-hand and right-hand sides represent the lowest and highest data outcomes since 2000, respectively. A pink dot to the right of a black dot means that the latest data outcome was stronger than in the December 2019 quarter.

5 Robinson (2020), 'Vacancies, unemployment and labour market slack in New Zealand', *Analytical Note*, Reserve Bank of New Zealand.

2

The neutral interest rate

The neutral interest rate is the short-term interest rate at which monetary policy is neither expansionary nor contractionary. It is determined by the long-run demand for capital investment and the supply of savings. The neutral interest rate influences the level that the OCR and other interest rates are expected to return to over the medium term, as the MPC meets its inflation and employment objectives.

The exact value of the neutral interest rate is unknown, as it cannot be measured directly. It also changes over time as factors that influence the long-term demand for capital investment and the supply of savings evolve. Nevertheless, the neutral interest rate is an important consideration when forecasting the economic outlook and setting monetary policy. It influences the broad level around which interest rates vary over the business cycle and the level we project the OCR to return to over time. We can also compare the OCR to estimates of the neutral rate to assess how stimulatory or contractionary monetary policy is at any given time.

The Reserve Bank uses a suite of indicators that combines several models and approaches to assess the neutral interest rate. Some of the indicators are based on expectations of interest rates over the long run, as well as other long-run economic factors that can influence the neutral interest rate. These indicators use expectations from professional forecasters and financial market participants, as implied by the prices of financial instruments (e.g. bond yields). Other indicators use information about recent economic outcomes, and consider what they imply about the level of monetary stimulus and the neutral interest rate.

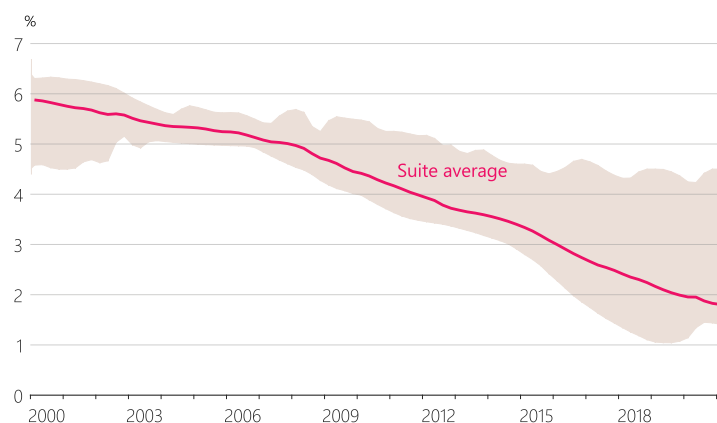
The OCR is below the neutral interest rate

Based on our suite of indicators, we estimate the nominal neutral interest rate to be about 2 percent, compared to our current OCR of 0.25 percent. While there is always a degree of uncertainty in the neutral interest rate estimate, this uncertainty – as reflected in the wide range of estimates from our suite – is heightened currently due to the impacts of the COVID-19 pandemic (figure 4.8). However, we are confident the current level of the OCR is below the neutral interest rate, consistent with the pace of the economic recovery and the OCR being below even the lowest indicator in the suite.

The MPC formulates monetary policy using a range of information and indicators. By combining several approaches, our indicator suite ensures that not too much emphasis is placed on any one model or framework. More importantly, the MPC regularly reviews economic outcomes and policy settings, which allows adjustments to be made if policy is not having the desired impact on the economy. For example, if the true neutral interest rate were lower than estimated, monetary policy would be less stimulatory than expected. Inflationary pressures and employment would be lower than anticipated and the MPC could respond by easing monetary policy if needed.

Figure 4.8

New Zealand neutral interest rate indicator suite (nominal)



Source: Bloomberg, Consensus Economics, Stats NZ and RBNZ estimates.

Global neutral rates have been trending down over time

Globally, estimates of neutral rates indicate they have been declining since at least the 1980s, a trend accelerated by the global financial crisis (GFC).⁶ The global decline in neutral interest rates reflects a number of factors that have reduced the long-term demand for capital investment or have increased the long-term supply of savings.

Improving life expectancy and decreasing fertility rates in developed countries have resulted in their populations ageing over the past few decades. This has led to a rise in the proportion of people who are at a stage in their lives when they are more likely to work and save, increasing the supply of savings globally.

Subdued productivity growth has also contributed by reducing returns on capital investment. A shift in the balance of growth towards less-capital-intensive industries (such as some services sectors) may have also contributed to lower demand for investment. These trends have been compounded by increased risk aversion and a desire among households and businesses to strengthen their balance sheets following the GFC.

Continued changes in demographics, risk preferences, and potential growth are expected to influence global neutral interest rates going forward. The implications of future demographic developments are complex, and may shift as more people retire and start drawing upon their savings.

For example, while a drain on savings would place upward pressure on the neutral rate, a declining labour force could result in an increased or decreased demand for investment. Assuming no technological advances, capital investment could decrease as fewer workers would need to be equipped. This would create downward pressure on the neutral rate, at least in the medium term. On the other hand, fewer workers, which increases the relative cost of labour, could encourage technological advancements to improve capital productivity. This would put upward pressure on the neutral interest rate.

Other factors that have led to a declining demand for capital investment and an increased supply of savings over past decades may also continue to influence the neutral rate. Overall, it is unclear whether global neutral interest rates will increase or decrease over time as a result of these trends, especially as some trends work in offsetting directions.

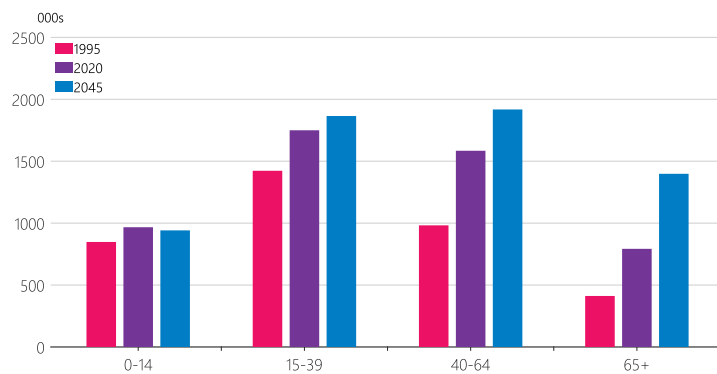
Global conditions contribute to declines in New Zealand's neutral interest rate

New Zealand is a small open economy with globally integrated financial markets. The global demand and supply of investment and savings, and by extension global neutral rates, have a large bearing on our neutral interest rate. In other words, New Zealand's neutral rate will be influenced by trends in global neutral rates, especially those of large economies such as the United States.

New Zealand has experienced many of the same trends that have reduced global neutral interest rates. Like in many other developed countries, New Zealand's population is ageing and is expected to do so over the next few decades. In 1995, 11 percent of the population was aged 65 years or over. This proportion is now around 16 percent, and is expected to increase to 23 percent by 2045 (figure 4.9). New Zealand has also seen declining productivity growth for some time (figure 4.10).

6 See online [New York Fed: Holston, Laubach, and Williams](#).

Figure 4.9
New Zealand's age profile



Source: Stats NZ.

Figure 4.10
Annual labour productivity growth



Source: Stats NZ.

The COVID-19 pandemic may influence the neutral rate

The COVID-19 pandemic has the potential to influence the neutral interest rate. Lockdowns and border restrictions have reduced the productive capacity of New Zealand. Likewise, heightened uncertainty at the beginning of the pandemic decreased the appetite for businesses to invest and encouraged households to save.

If sustained in the long term, these types of change would typically place downward pressure on the neutral rate. However, these drivers are expected to be largely temporary, subsiding as border restrictions ease and New Zealand continues its strong economic recovery. As these factors are not expected to cause significant longer-term changes in the behaviour of businesses, households, or the structure of the economy, we do not believe they will have a significant impact on the neutral interest rate.

However, the pandemic may also have some longer-lasting impacts that could influence the level of the neutral interest rate. For example, the pandemic and the associated global fiscal responses have had a significant bearing on household, business, and government balance sheets, including in New Zealand. At the moment, it is not clear whether these factors will have net upward or downward impacts on the neutral interest rate.



Economic
projection

CHAPTER
05

CHAPTER 5

Economic projection

This chapter summarises the baseline economic projection that the MPC considered when making its policy assessment. The projection was finalised on 18 August 2021. The OCR projection incorporates an outlook for monetary policy that is consistent with the MPC’s monetary policy assessment, which was finalised on 18 August. Other data and developments have been incorporated up until 11 August.

This projection relies on a set of key assumptions about the global and domestic responses to the COVID-19 pandemic and the subsequent recovery. Substantially different outcomes could eventuate if these assumptions do not hold. Significant uncertainty surrounds our assumptions about the effectiveness of vaccine programmes both globally and domestically, the spread of highly contagious variants, and their impacts on economic activity. There is also uncertainty about how the behaviour of businesses and consumers will change relative to pre-COVID-19 times and the effects of significant changes in fiscal policies in New Zealand and abroad.

The projection takes into account recent data, which show a strong rebound in economic activity in early 2021 and a more resilient labour market than previously anticipated. The factors underpinning this starting point are discussed in chapters 2–4. The projection also accounts for the assumed impacts of developments in New Zealand and abroad since the May *Statement*, including the announced pause of the trans-Tasman bubble. The projection assumes that New Zealand is predominantly at Alert Level 1 or a lower level of restrictions. The majority of New Zealand’s adult population is assumed to be fully vaccinated by the end of 2021. Although there remains

a risk of widespread COVID-19 outbreaks in New Zealand, we assume outbreaks will be sporadic and quickly contained. Border restrictions are assumed to ease more gradually from the beginning of 2022, in line with recent Government announcements. Employment rises above its maximum sustainable level (MSE) over the forecast horizon due to intensifying capacity pressures, but returns towards MSE by the end of the projection as pressures ease. Inflation remains above the 2 percent target mid-point over the projection due to sustained capacity pressures. Inflation is boosted by near-term supply shocks that are assumed to be transitory.

The economic outlook means that monetary stimulus is projected to be reduced over the forecast horizon to ensure the MPC meets its medium-term inflation and employment objectives.

Chapter 2 discusses some of the key risks to this economic outlook. These include:

- COVID-19 resurgence in New Zealand;
- higher inflation becoming embedded;
- the economic effects of easing border restrictions; and
- uncertainty regarding the longer-term economic implications of COVID-19.

Table 5.1

Key projection assumptions

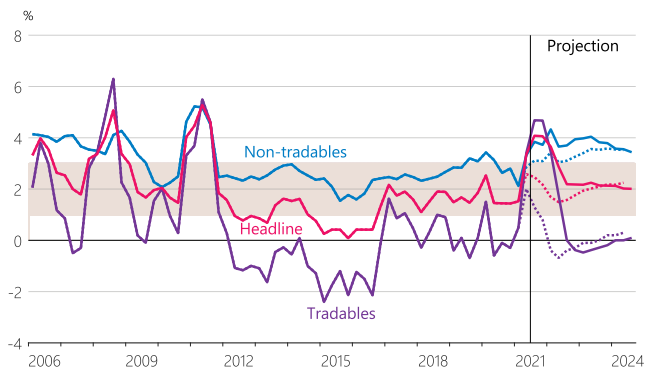
Key factors	Global growth	• The annual average GDP growth of our trading partners is projected to recover to 6.6% in 2021 and moderate thereafter. The global economic recovery is being supported by the ongoing vaccine rollout. • The New Zealand dollar trade-weighted index (TWI) is assumed to stay around 74.4, above its pre-COVID-19 level. • The Dubai oil price is assumed to return gradually to around 65 USD per barrel over the medium term, in line with futures market pricing.
	Health restrictions	• New Zealand is assumed to predominantly be at Alert Level 1 or a lower level of restrictions over the projection. Any COVID-19 outbreaks are assumed to be sporadic and quickly contained. • The small but positive anticipated impact of the trans-Tasman travel bubble on net exports over the projection is assumed to be lower due to the suspension of the bubble in the September quarter. If the bubble re-starts, uptake is expected to be fairly gradual, hampered by concerns over its ongoing stability. • A significant proportion of New Zealand's adult population is assumed to be vaccinated by the end of 2021. Wider border restrictions are expected to begin to ease gradually in 2022, in line with recent Government announcements.
	House prices	• Annual average house price growth is assumed to decrease over the projection period, from 26.4% in 2021 to 7.9% in 2022. House prices are assumed to decline modestly from late 2022 until the end of the forecast horizon. This reflects a variety of factors, including increases in interest rates, increased housing supply, low net immigration, announced changes to tax policy, and the reintroduction and tightening of LVR restrictions.

Economic growth	Production	- Quarterly GDP growth is expected to ease in mid-2021 as services growth moderates from its current high level. Annual growth rises to 4.0% in the December 2021 quarter, supported by strong household income growth and some residual momentum in the housing market. - New Zealand's productive capacity (potential output) has fallen as a result of the domestic and international measures put in place to contain COVID-19. Mismatches in labour demand and supply, across skills and regions, have resulted in slightly higher structural unemployment. These temporary disruptions are expected to dissipate as domestic border restrictions ease. - The output gap is starting from a much higher position than assumed in the <i>May Statement</i>. This reflects stronger-than-expected economic growth, a tighter labour market, and surveys of businesses that show increased capacity pressures.
	Consumption	- Consumption is expected to soften in the June 2021 quarter, largely reflecting a reversal of higher-than-usual domestic spending during the summer. - Nevertheless, consumption is projected to be supported in the near term by a more resilient labour market, savings accumulated during 2020, and the impacts of recent strong house price inflation. The impact of higher interest rates is anticipated to provide some offset, particularly over the medium term. - Consumption remains above pre-COVID-19 levels over the projection horizon.
	Investment	- Business investment is expected to remain above pre-COVID-19 levels over the projection horizon due to higher demand, lower uncertainty, and increased capacity pressures. - Residential investment peaks in the September 2021 quarter, reflecting strong growth in residential building consent issuance. Residential investment is expected to remain elevated over coming quarters. - Residential investment is assumed to then soften somewhat from the end of 2021 as house price inflation eases. Nevertheless, easier urban planning rules are assumed to support residential investment over the medium term.
	Exports and imports	- Improving global demand is expected to underpin demand for New Zealand's exports. This supports stronger, albeit moderating, export prices over the projection. - Import prices are higher than previously expected, largely driven by higher oil prices and the impact of supply-chain disruptions on the cost of goods and services. - The terms of trade are slightly lower than expected in the <i>May Statement</i>. The current account balance is assumed to be lower due to higher-than-expected import prices and volumes, which are partially offset by higher export prices.

Labour market	Employment and wages	• The unemployment rate has peaked at 5.3% in the September 2020 quarter. Over the projection, the unemployment rate is assumed to trough at 3.8% in the September 2022 quarter as economic activity recovers and capacity pressures remain strong. • Structural unemployment is assumed to be elevated relative to pre-COVID-19 levels, largely reflecting a skills mismatch in the New Zealand labour market. This has lowered the level of maximum sustainable employment. Currently, employment is estimated to be at or above its maximum sustainable level. • Annual LCI private sector wage inflation is assumed to accelerate from 2.2% in the June 2021 quarter to 2.7% at the end of the year. This reflects a tight labour market. Annual wage inflation peaks at 2.9% later in the projection as capacity pressures tighten. • Net working-age immigration is projected to be around 1,500 people over 2021, with the assumption that the trans-Tasman bubble has no significant impact on the net flow of migrants. Net immigration is assumed to increase gradually as border restrictions ease, to a long-run average of about 24,000 people per year.
Inflation	Headline	• Annual CPI inflation rises above the 1–3% target band in the near term, but declines to the 2% midpoint at the end of the projection period. • Headline inflation is projected to peak at around 4% in the September 2021 quarter, but some of the factors lifting inflation are assumed to be relatively short-lived. Headline inflation is expected to stay slightly above 2% over most of the remainder of the forecast horizon due to sustained capacity pressures. • Supply-chain and transportation disruptions are anticipated to remain over much of 2022, longer than assumed in the <i>May Statement</i>.
	Tradables	• Annual tradables inflation is expected to increase temporarily above 4% in late 2021, largely due to recent increases in oil prices and the ongoing impact of supply-chain disruptions. • Over the medium term, moderating oil prices and a relatively strong TWI keep annual tradables inflation muted. Annual tradables inflation reaches around 0% by the end of the projection.
	Non-tradables	• Annual non-tradables inflation remains at or above 3.4% over the projection. • High starting-point momentum in house prices and construction costs supports housing-related components of non-tradables inflation in the near and medium term. Higher council rates also boost non-tradables inflation in the near term. Over the projection, sustained capacity pressures, including a tight labour market, provide support as transitory factors dissipate.

Charts

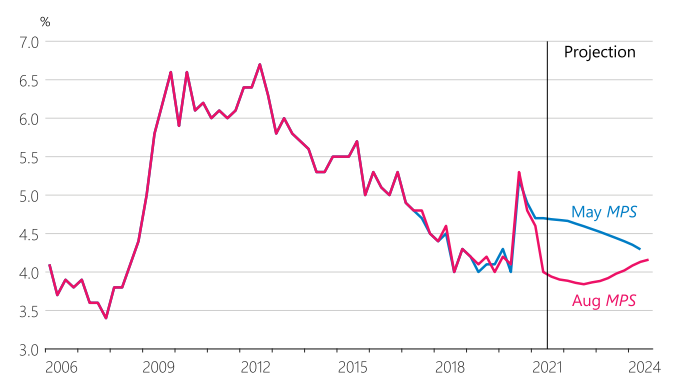
Figure 5.1
Inflation breakdown
(annual)



Source: Stats NZ, RBNZ estimates.

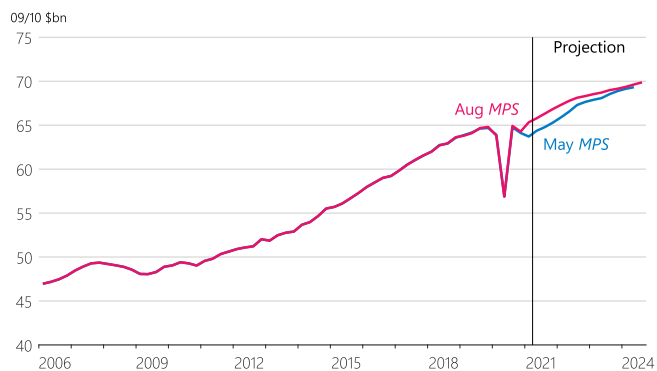
Note: Dotted lines show the projection from the May Statement.

Figure 5.3
Unemployment rate
(seasonally adjusted)



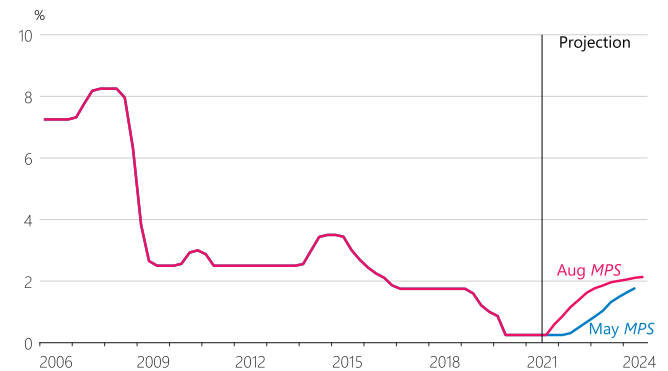
Source: Stats NZ, RBNZ estimates.

Figure 5.2
Quarterly production GDP
(seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Figure 5.4
OCR



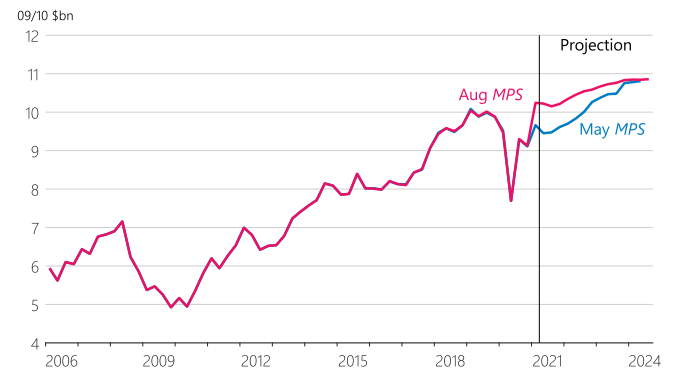
Source: RBNZ estimates.

Figure 5.5
Terms of trade



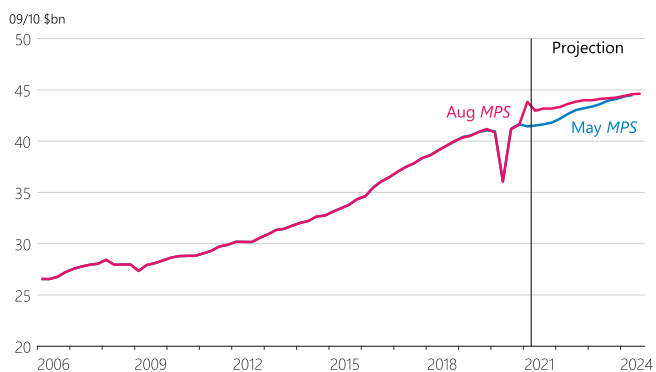
Source: Stats NZ, RBNZ estimates.

Figure 5.7
Business investment
(seasonally adjusted)



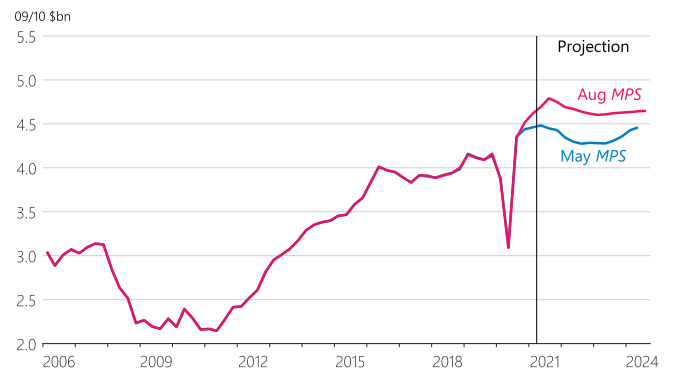
Source: Stats NZ, RBNZ estimates.

Figure 5.6
Household consumption
(seasonally adjusted)



Source: Stats NZ, RBNZ estimates.

Figure 5.8
Residential investment
(seasonally adjusted)



Source: Stats NZ, RBNZ estimates.



Appendices

CHAPTER 06

CHAPTER 6

Appendices

Appendix 1: Statistical tables

Table 6.1

Key forecast variables

		GDP growth Quarterly	CPI inflation Quarterly	CPI inflation Annual	TWI	OCR
2019	Mar	0.3	0.1	1.5	74.0	1.8
	Jun	0.4	0.6	1.7	72.6	1.6
	Sep	0.9	0.7	1.5	72.0	1.2
	Dec	0.2	0.5	1.9	71.3	1.0
2020	Mar	-1.5	0.8	2.5	70.9	0.9
	Jun	-10.8	-0.5	1.5	69.7	0.3
	Sep	14.1	0.7	1.4	72.0	0.3
	Dec	-1.0	0.5	1.4	72.9	0.3
2021	Mar	1.6	0.8	1.5	74.9	0.3
	Jun	0.7	1.3	3.3	74.7	0.3
	Sep	0.8	1.4	4.1	74.4	0.3
	Dec	0.8	0.5	4.1	74.4	0.6
2022	Mar	0.7	0.5	3.7	74.4	0.9
	Jun	0.7	0.5	2.9	74.4	1.2
	Sep	0.5	0.7	2.2	74.4	1.4
	Dec	0.3	0.4	2.2	74.4	1.6
2023	Mar	0.3	0.4	2.2	74.4	1.8
	Jun	0.3	0.6	2.2	74.4	1.9
	Sep	0.4	0.6	2.1	74.4	2.0
	Dec	0.2	0.4	2.1	74.4	2.0
2024	Mar	0.3	0.4	2.1	74.4	2.1
	Jun	0.4	0.5	2.0	74.4	2.1
	Sep	0.4	0.6	2.0	74.4	2.1

Table 6.2

Measures of inflation, inflation expectations, and asset prices

	2020				2021			
	Dec	Mar	Jun	Sep	Dec	Mar	Jun	Sep
Inflation (annual rates)								
CPI	1.9	2.5	1.5	1.4	1.4	1.5	3.3	
CPI non-tradables	3.1	3.4	3.1	2.6	2.8	2.1	3.3	
CPI tradables	0.1	1.5	-0.6	-0.1	-0.3	0.5	3.4	
Sectoral factor model estimate of core inflation	1.8	1.8	1.8	1.8	1.9	2.0	2.2	
CPI trimmed mean (30 percent measure)	2.1	2.7	2.1	1.7	2.2	1.7	3.0	
CPI weighted median	2.3	2.8	2.4	2.2	2.7	2.3	3.0	
GDP deflator (expenditure)	3.6	3.7	2.5	1.6	0.7	0.6		
Inflation expectations								
ANZ Business Outlook – inflation one year ahead (quarterly average to date)	1.7	1.7	1.4	1.4	1.5	1.9	2.2	2.7
RBNZ Survey of Expectations – inflation 2 years ahead	1.8	1.9	1.2	1.4	1.6	1.9	2.0	2.3
RBNZ Survey of Expectations – inflation 5 years ahead	2.0	2.0	1.8	1.9	1.9	2.0	2.1	2.0
RBNZ Survey of Expectations – inflation 10 years ahead	2.0	2.0	2.0	2.0	2.1	2.0	2.0	2.0
Long-run inflation expectations*	2.0	1.8	1.8	2.0	2.0	2.1	2.1	2.1
Asset prices (annual percent changes)								
Quarterly house price index (CoreLogic NZ)	4.5	7.7	7.1	10.8	16.9			
REINZ Farm Price Index (quarterly average)	-0.8	-12.5	-8.0	-3.7	-0.5	6.9	6.3	
NZX 50 (quarterly average)	25.7	21.1	6.4	8.6	13.2	14.4	17.1	

*Long-run expectations are extracted from a range of surveys using a Nelson-Siegel model. Source: ANZ Bank, Aon-Hewitt Consulting, Consensus Economics, RBNZ estimates.

Table 6.3

Measures of labour market conditions*(seasonally adjusted, changes expressed in annual percent terms)*

	2020				2021		
	Dec	Mar	Jun	Sep	Dec	Mar	Jun
Household Labour Force Survey							
Unemployment rate	4.0	4.2	4.1	5.3	4.8	4.6	4.0
Underutilisation rate	10.1	10.4	12.1	13.1	11.8	12.1	10.5
Labour force participation rate	70.4	70.7	69.9	70.1	70.2	70.4	70.5
Employment rate (percentage of working-age population)	67.6	67.7	67.0	66.4	66.8	67.1	67.7
Employment growth	1.2	2.5	1.6	0.4	0.7	0.3	1.7
Average weekly hours worked	33.8	34.1	30.7	33.7	34.9	33.9	34.1
Number unemployed (thousand people)	114	120	117	151	139	133	117
Number employed (million people)	2.72	2.74	2.73	2.72	2.73	2.75	2.78
Labour force (million people)	2.83	2.86	2.85	2.87	2.87	2.88	2.89
Extended labour force (million people)	2.92	2.96	2.96	2.97	2.97	2.99	2.99
Working-age population (million people)	4.02	4.05	4.08	4.09	4.09	4.10	4.11
Quarterly Employment Survey – QEM							
Filled jobs growth	2.9	2.1	0.3	-0.6	0.3	0.5	2.4
Average hourly earnings growth (private sector, ordinary time)	2.5	3.2	2.4	3.7	4.6	4.1	4.4
Other data sources							
Labour cost index growth, private sector	2.4	2.4	1.7	1.6	1.5	1.6	2.2
Labour cost index growth, private sector, unadjusted	3.7	3.6	2.8	2.6	2.4	2.8	3.7
Estimated net migration (published, thousands, quarterly)	21.4						
Change in All Vacancies Index	-4.8	-13.3	-49.8	-19.7	1.7	23.0	166.9

Note: The All Vacancies Index is produced by MBIE as part of the monthly Jobs Online report, which shows changes in job vacancies advertised by businesses on internet job boards. It is converted to quarterly and seasonally adjusted by the RBNZ. The unadjusted labour cost index (LCI) is an analytical index that reflects quality change in addition to price change (whereas the official LCI measures price changes only). For definitions of underutilisation, the extended labour force, and related concepts, see Statistics New Zealand (2016), *Introducing underutilisation in the labour market*. Estimated net migration (published) is the Stats NZ outcomes-based measure.

Table 6.4**Composition of real GDP growth***(annual average percent change, seasonally adjusted, March years, unless specified otherwise)*

March year	Actuals								Projection		
	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Final consumption expenditure											
Private	3.9	3.3	4.2	6.5	4.9	4.4	2.8	-0.5	6.2	1.6	0.9
Public authority	2.1	3.4	2.3	2.2	3.4	3.7	6.1	6.2	4.4	1.4	0.7
Total	3.5	3.3	3.7	5.5	4.6	4.2	3.6	1.1	5.7	1.6	0.8
Gross fixed capital formation											
Residential	15.2	8.3	7.1	8.8	-1.8	3.0	1.5	2.0	14.2	-2.1	-0.2
Other	7.6	7.9	2.8	0.3	10.7	6.6	1.1	-6.7	12.9	4.3	1.8
Total	9.4	8.0	3.9	2.5	7.3	5.7	1.2	-4.6	13.2	2.6	1.3
Final domestic expenditure	4.8	4.4	3.8	4.8	5.2	4.6	3.0	-0.2	7.4	1.8	0.9
Stockbuilding*	-0.2	0.5	-0.3	0.1	0.2	-0.1	-0.5	-0.2	0.8	-0.3	0.0
Gross national expenditure	4.5	4.6	3.3	5.0	5.7	4.5	2.4	-1.3	8.3	2.0	1.1
Exports of goods and services	0.1	4.7	6.6	1.7	3.6	3.2	-0.3	-15.7	5.6	7.7	5.5
Imports of goods and services	8.1	7.7	2.6	5.6	7.8	4.4	1.0	-16.1	16.8	5.4	4.1
Expenditure on GDP	2.1	3.7	4.4	3.8	4.4	4.1	2.1	-0.4	5.2	2.3	1.2
GDP (production)	2.7	3.8	3.7	3.7	3.6	3.3	1.7	-2.3	5.9	2.4	1.3
GDP (production, March qtr to March qtr)	3.5	3.8	4.1	3.2	3.6	2.9	0.0	2.4	3.0	1.8	1.2

*Percentage point contribution to the growth rate of GDP.

Table 6.5

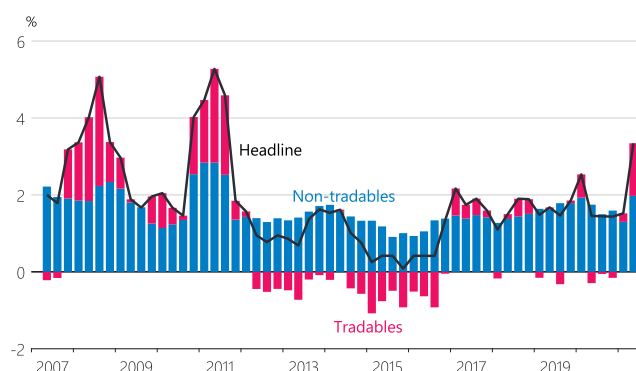
Summary of economic projection

(annual percent change for March years unless specified otherwise)

March year	Actuals								Projection		
	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Price measures											
CPI	1.5	0.3	0.4	2.2	1.1	1.5	2.5	1.5	3.7	2.2	2.1
Labour costs	1.7	1.8	1.8	1.5	1.9	2.0	2.4	1.6	2.9	2.9	2.7
Export prices (in New Zealand dollars)	11.5	-9.2	-0.3	3.9	3.2	1.3	7.3	-3.2	4.2	-2.9	-0.2
Import prices (in New Zealand dollars)	-3.0	-3.4	1.2	0.6	1.6	4.0	2.3	-3.2	2.5	-1.6	0.3
Monetary conditions											
OCR (year average)	2.5	3.4	2.9	2.0	1.8	1.8	1.2	0.3	0.5	1.5	2.0
TWI (year average)	77.6	79.3	72.6	76.5	75.6	73.4	71.7	72.4	74.5	74.4	74.4
Output											
GDP (production, annual average % change)	2.7	3.8	3.7	3.7	3.6	3.3	1.7	-2.3	5.9	2.4	1.3
Potential output (annual average % change)	2.6	3.1	3.2	3.3	3.1	3.0	2.5	-2.3	4.6	2.0	1.9
Output gap (% of potential GDP, year average)	-1.5	-0.8	-0.3	0.2	0.6	0.8	0.1	0.0	1.2	1.6	1.0
Labour market											
Total employment (seasonally adjusted)	4.0	3.6	2.3	5.9	2.9	1.5	2.5	0.3	1.8	1.0	1.0
Unemployment rate (March qtr, seasonally adjusted)	5.6	5.5	5.3	4.9	4.4	4.2	4.2	4.6	3.9	3.9	4.1
Trend labour productivity	0.8	0.7	0.6	0.4	0.3	0.2	0.3	0.5	0.7	0.7	0.7
Key balances											
Government operating balance (% of GDP, year to June)	-1.2	0.2	0.7	1.5	1.9	2.4	-7.3	-4.3	-4.5	-2.6	-2.2
Current account balance (% of GDP)	-2.5	-3.5	-2.5	-2.7	-3.2	-3.9	-2.8	-2.2	-4.0	-4.2	-4.5
Terms of trade (SNA measure, annual average % change)	11.7	-0.3	-3.0	2.6	4.6	-2.5	2.0	-0.7	3.6	-0.9	-1.2
Household saving rate (% of disposable income)	1.2	-0.3	0.0	0.3	-0.1	0.5	1.9	5.5	2.4	1.4	0.6
World economy											
Trading-partner GDP (annual average % change)	3.5	3.7	3.5	3.5	3.9	3.5	1.6	-0.9	6.5	4.3	3.4
Trading-partner CPI (TWI weighted)	2.2	1.0	1.2	1.9	1.8	1.4	2.4	0.8	2.4	1.9	2.2

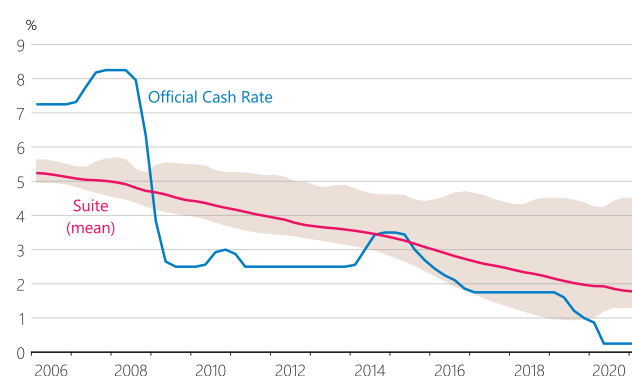
Appendix 2: Chart pack

Figure 6.1
Composition of CPI inflation
(annual)



Source: Stats NZ, RBNZ estimates.

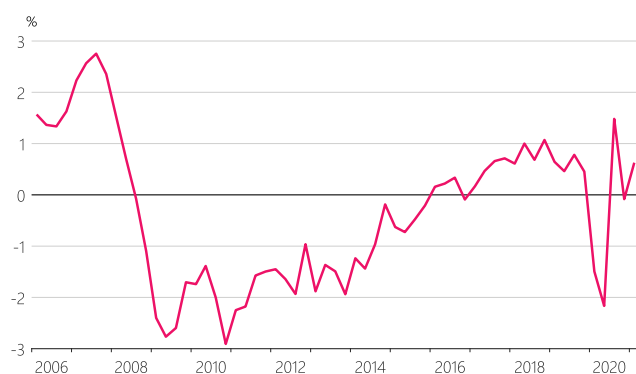
Figure 6.4
OCR and neutral OCR indicator suite
(quarterly average)



Source: RBNZ estimates.

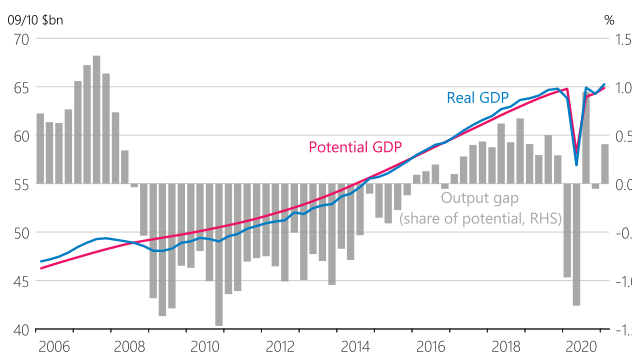
Note: Shaded area indicates the range between the maximum and minimum values from a suite of neutral OCR indicators.

Figure 6.2
Output gap
(share of potential)



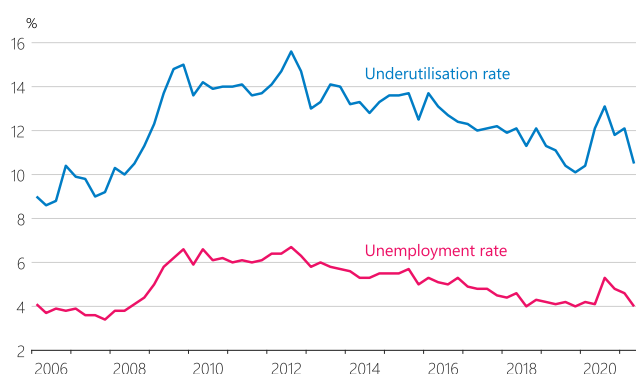
Source: Stats NZ, RBNZ estimates.

Figure 6.5
GDP and potential GDP
(seasonally adjusted)



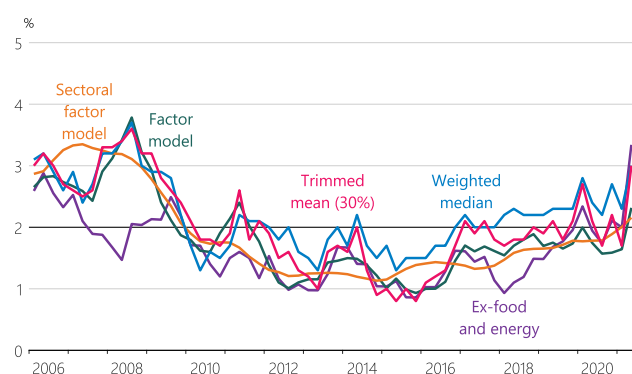
Source: Stats NZ, RBNZ estimates.

Figure 6.3
Unemployment and underutilisation rates
(seasonally adjusted)



Source: Stats NZ.

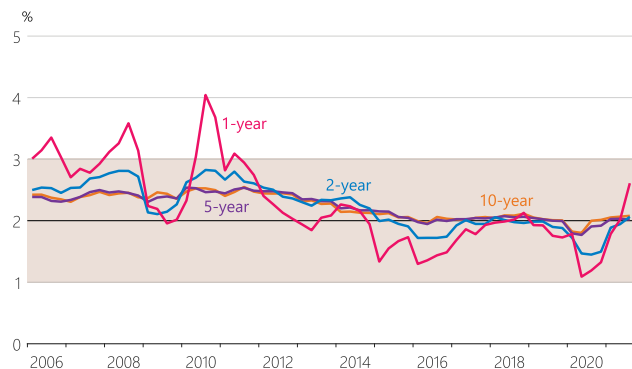
Figure 6.6
Headline inflation and core inflation
(annual)



Source: Stats NZ, RBNZ estimates.

Note: Core inflation measures exclude the GST increase in 2010.

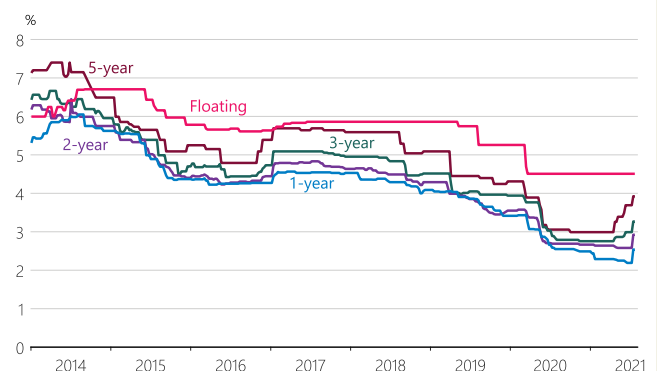
Figure 6.7
Inflation expectations
(annual)



Source: RBNZ estimates.

Note: Inflation expectations are estimates from the RBNZ inflation expectations curve, based on surveys of businesses and professional forecasters.

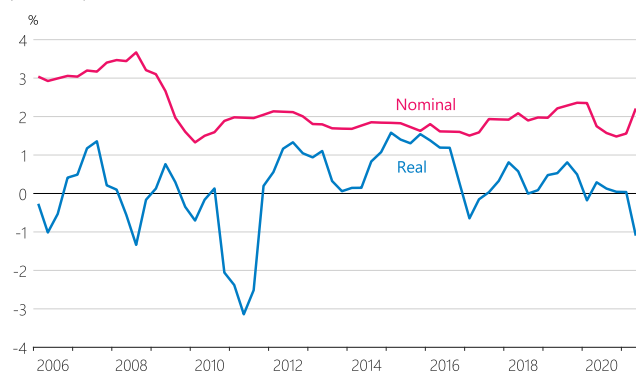
Figure 6.10
Mortgage rates



Source: interest.co.nz, RBNZ estimates.

Note: The rates shown for each term are the average of the latest rates on offer from ANZ, ASB, BNZ, and Westpac.

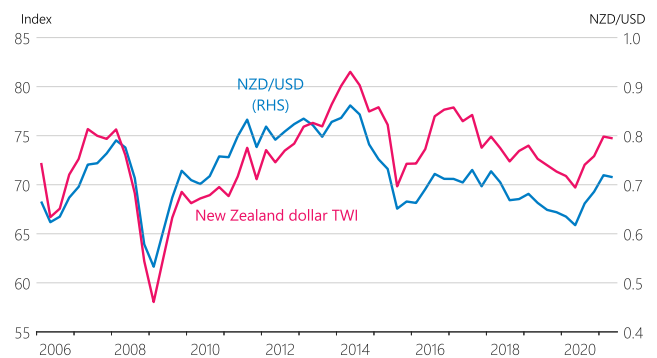
Figure 6.8
Private sector wage growth
(annual)



Source: Stats NZ, RBNZ estimates.

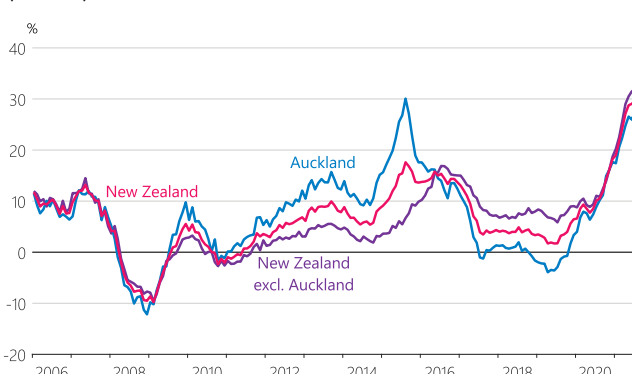
Note: Private sector wage growth is measured by the labour cost index, all salary and wage rates, private sector. Real labour cost index is deflated with headline CPI inflation.

Figure 6.11
New Zealand dollar exchange rates
(quarterly average)



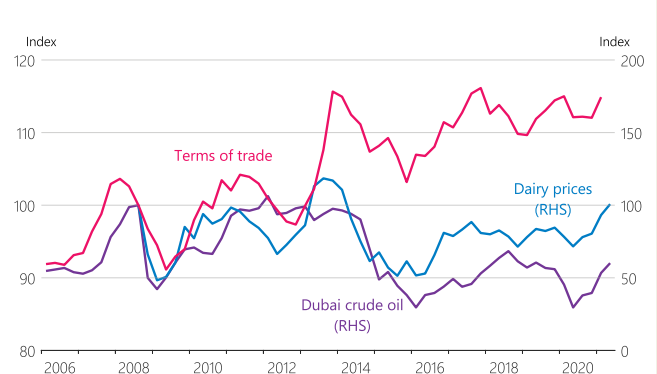
Source: Reuters, RBNZ estimates.

Figure 6.9
House price inflation
(annual)



Source: REINZ.

Figure 6.12
Terms of trade, dairy and oil price indices



Source: Stats NZ, Global Dairy Trade, Reuters, RBNZ estimates.